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基于STM32G4电机控制 基础篇

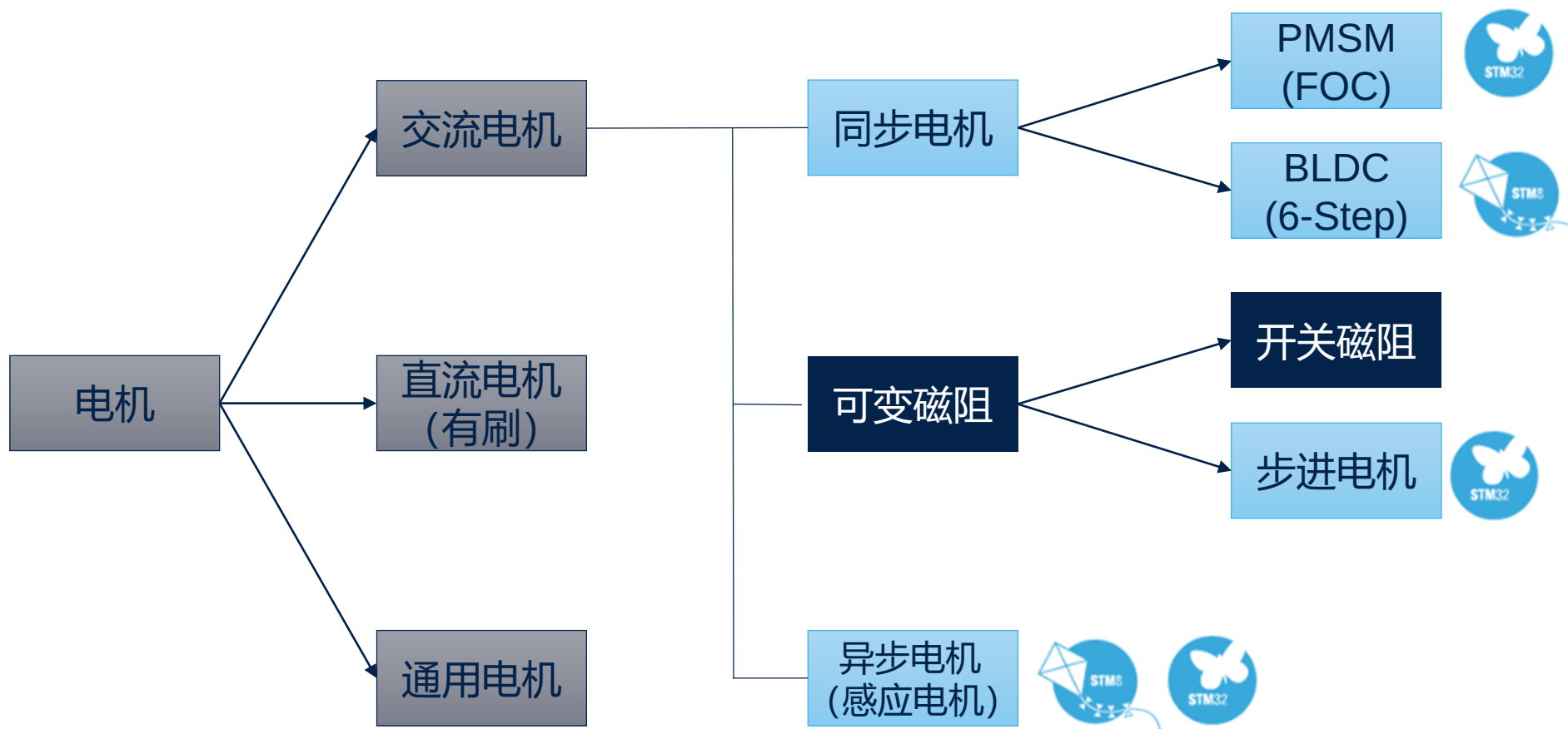
意法半导体微控制器部门

Agenda

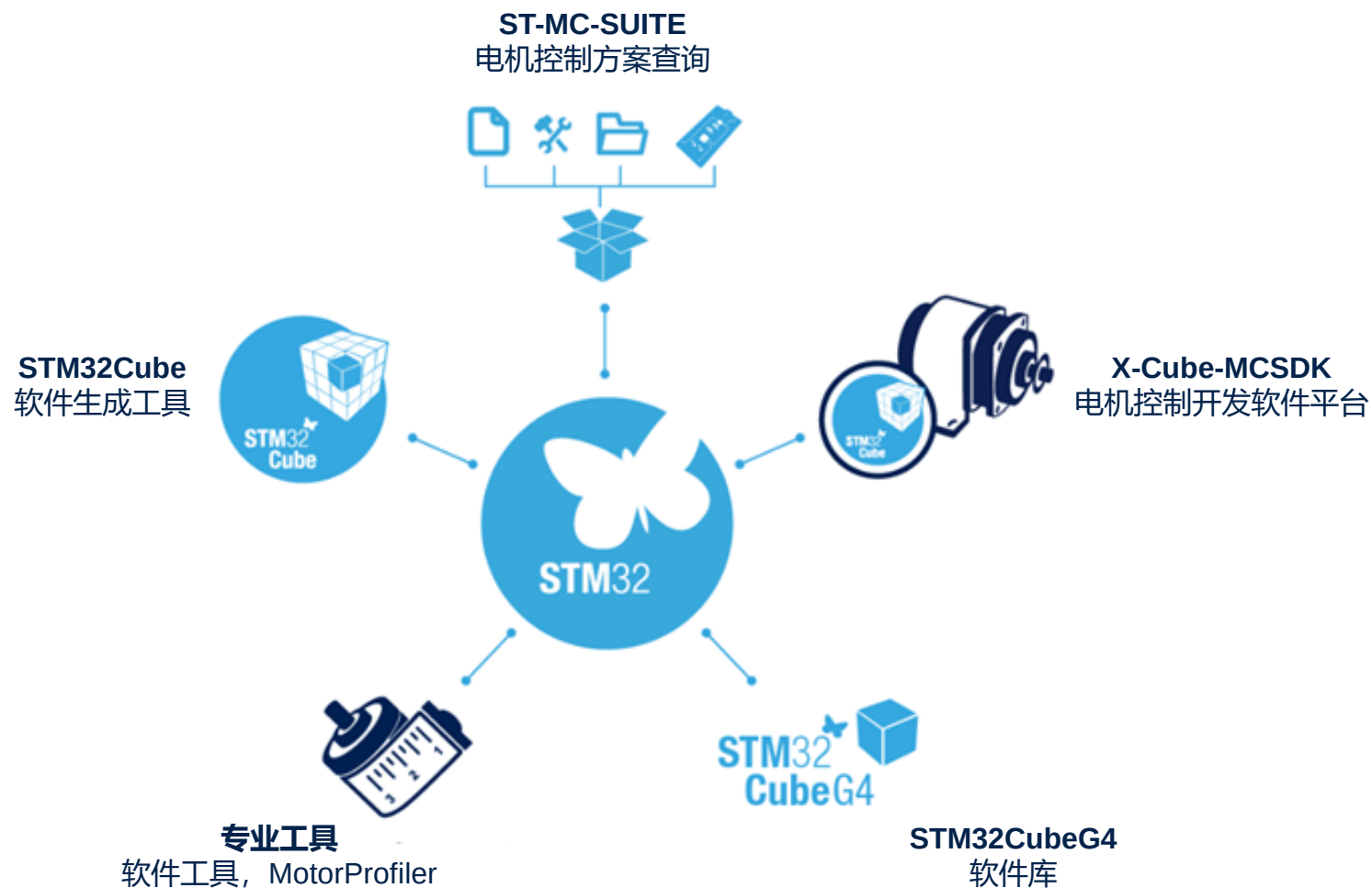
- 1 ST电机控制生态
- 2 MC SDK生成STM32G4电机FOC代码
- 3 基于CubeMx移植电机控制代码

STM32G4电机控制生态

电机及控制



STM32G4电机控制生态



有用的网站

- 电机培训资料下载: www.stmcu.com.cn
- 电机控制方案查询: <https://www.st.com/st-mc-suite/home/#/index>
- 电机培训视频: AI电堂 (微信)

设计资源分类

STM32 STM8 应用

全部 云接入 以太网 马达 USB GUI RTOS 其他

芯片文档 固件和软件 评估开发板

全部 中文译文 实战经验 培训课件及视频

培训课件及视频 (8)

培训名称	课件 (文件大小)	下载次数	视频 (文件大小)
STM32电机控制应用系列讲座之一...	(0.1M)	178	
文档说明: STM32电机控制应用系列讲座之一, 主要介绍了ST MC SDK5.x是什么, 支持的MCU, MotorControlWorkBench, 电动机参数测试, 以及参考资料列表。			
STM32电机控制应用系列讲座之二...	(0.1M)	103	
文档说明: STM32电机控制应用系列讲座之二, 主要介绍了永磁同步电动机的转动原理, 能量变...			

life.augmented STM32 STM8 Motor Control Suite

Browse motor control resources

Create and download a bundle

Save motor control setups

STM32电机控制应用系列讲座

课程主题:

- ST MC SDK5.x概览
- 永磁同步电动机矢量控制基础
- 电动机相电流检测与重构方法及转子位置检测与估计方法
- ST MC SDK 5.2 WB应用指南及ST MC SDK5.2 固件详解
- 应用ST MC SDK5.2 及 ST 硬件评价板调试电机实例
- ST MC SDK5.2 电动机参数测试

STM32电机控制应用系列讲座

本课程旨在讲解 ST MCU 在电机控制领域的应用。

已更新24期 | 147人订阅

展开描述

内容

电机控制答疑【回放二】

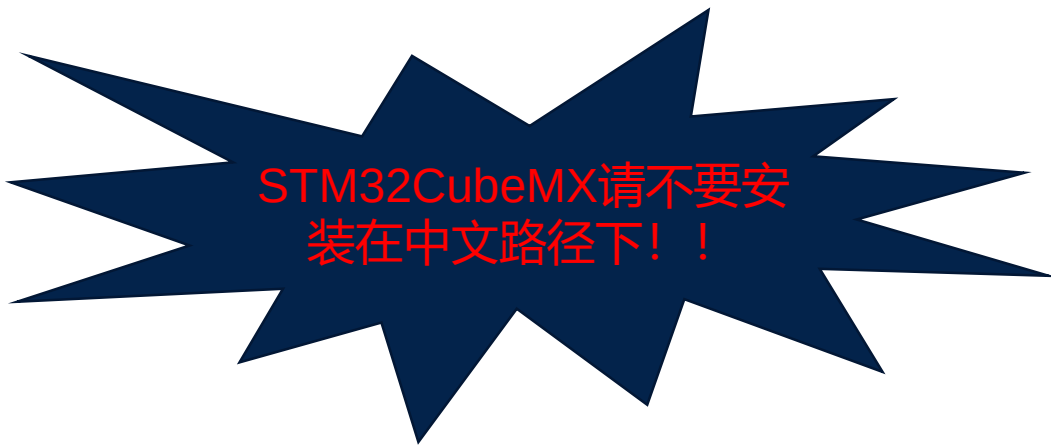
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2019-03-11

MC SDK生成STM32G4电机FOC代码

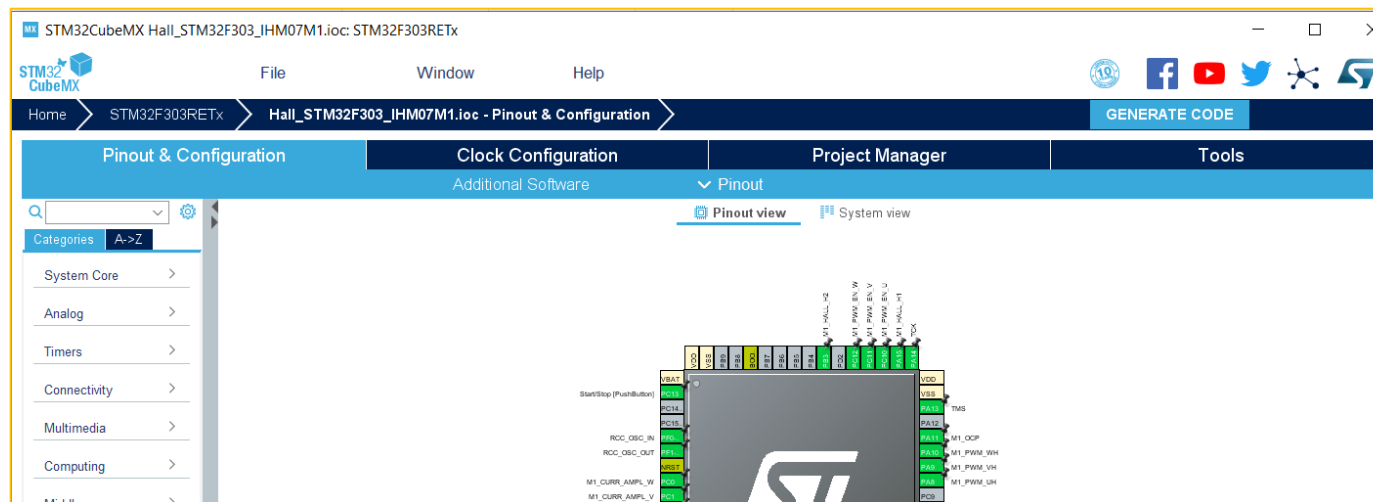
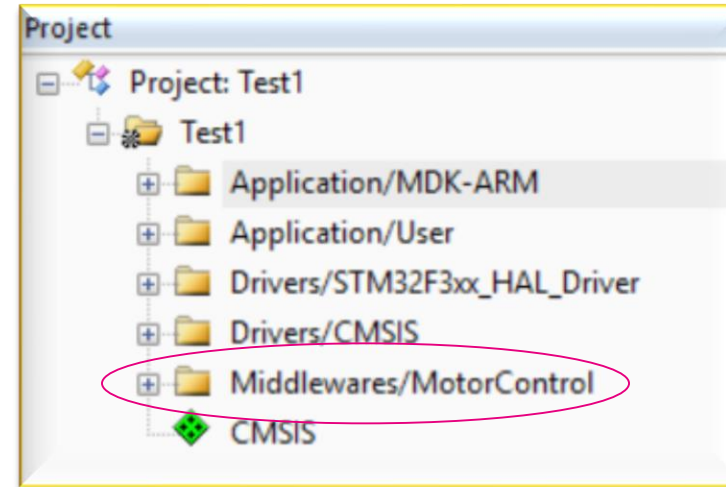
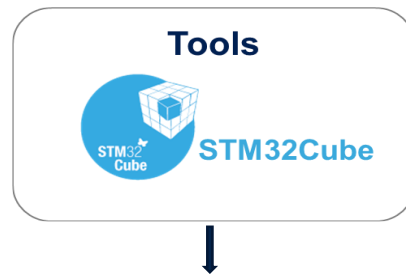
软件工具安装

- 请预先安装下列PC软件工具：
 - X-CUBE-MCSDK 或 X-CUBE-MCSDK-FUL
 - STM32CubeMX (v5.6以上)及**固件库**
 - ST-LINK/V2 或者ST-LINK/V3
 - IDE:
 - IAR Embedded Workbench for Arm (v8.x)
 - µVision® IDE for Arm® (Keil® MDK) v5.x
 - Atollic TrueSTUDIO for STM32 version 9.0.0



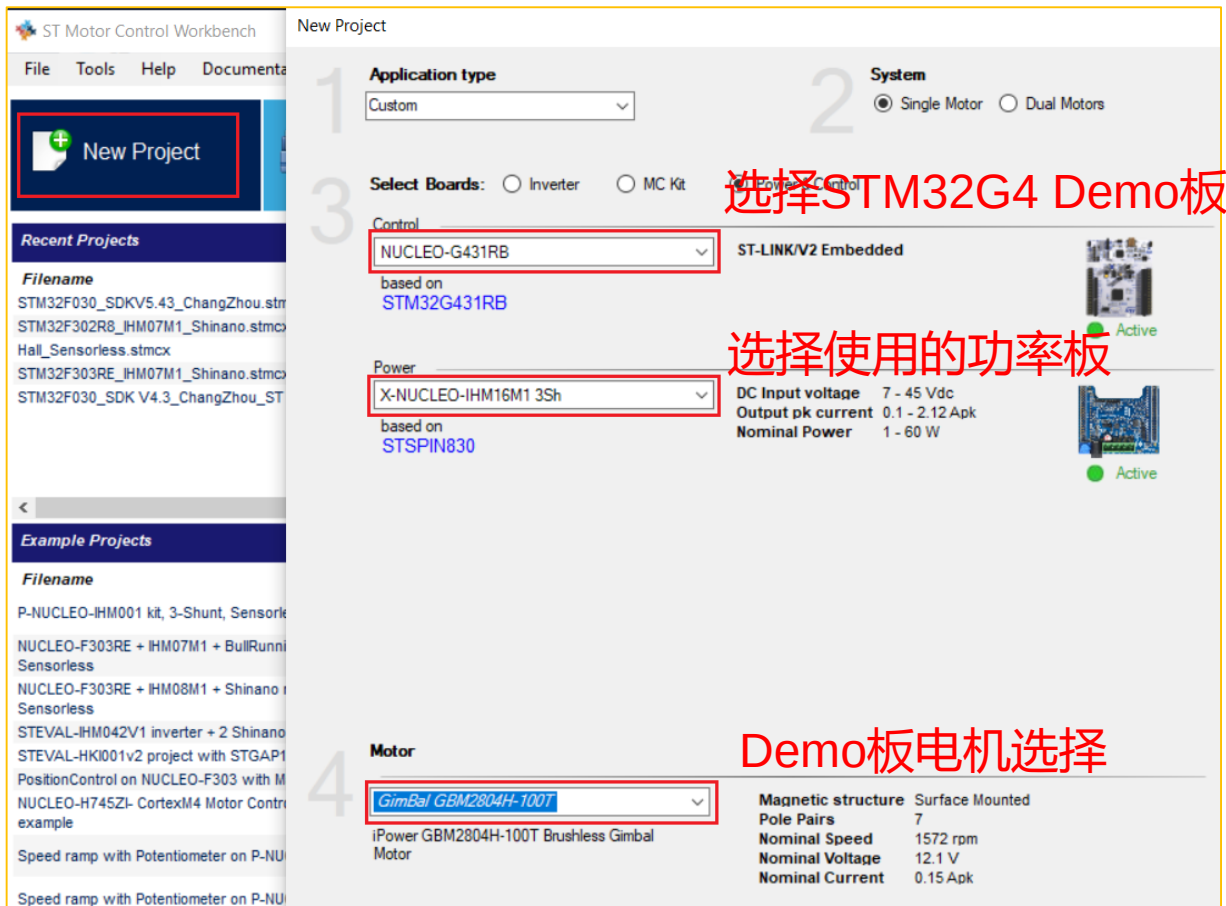
STM32CubeMX请不要安装在中文路径下！！

MC SDK5.x电机控制代码自动生成



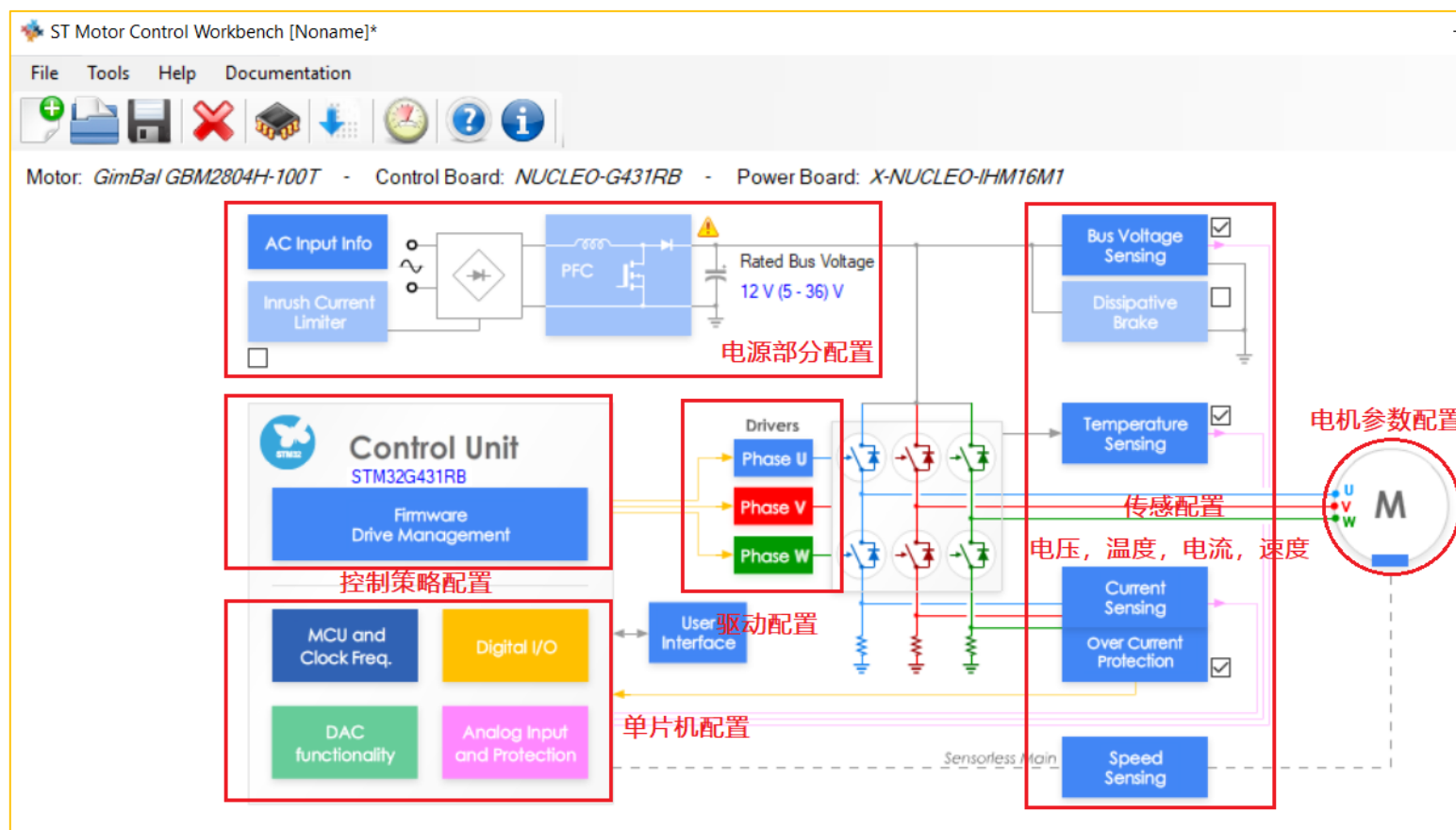
Workbench创建工程

- 本例中使用电机控制套件P-NUCLEO-IHM03
- 可在Workbench中进行对应选择

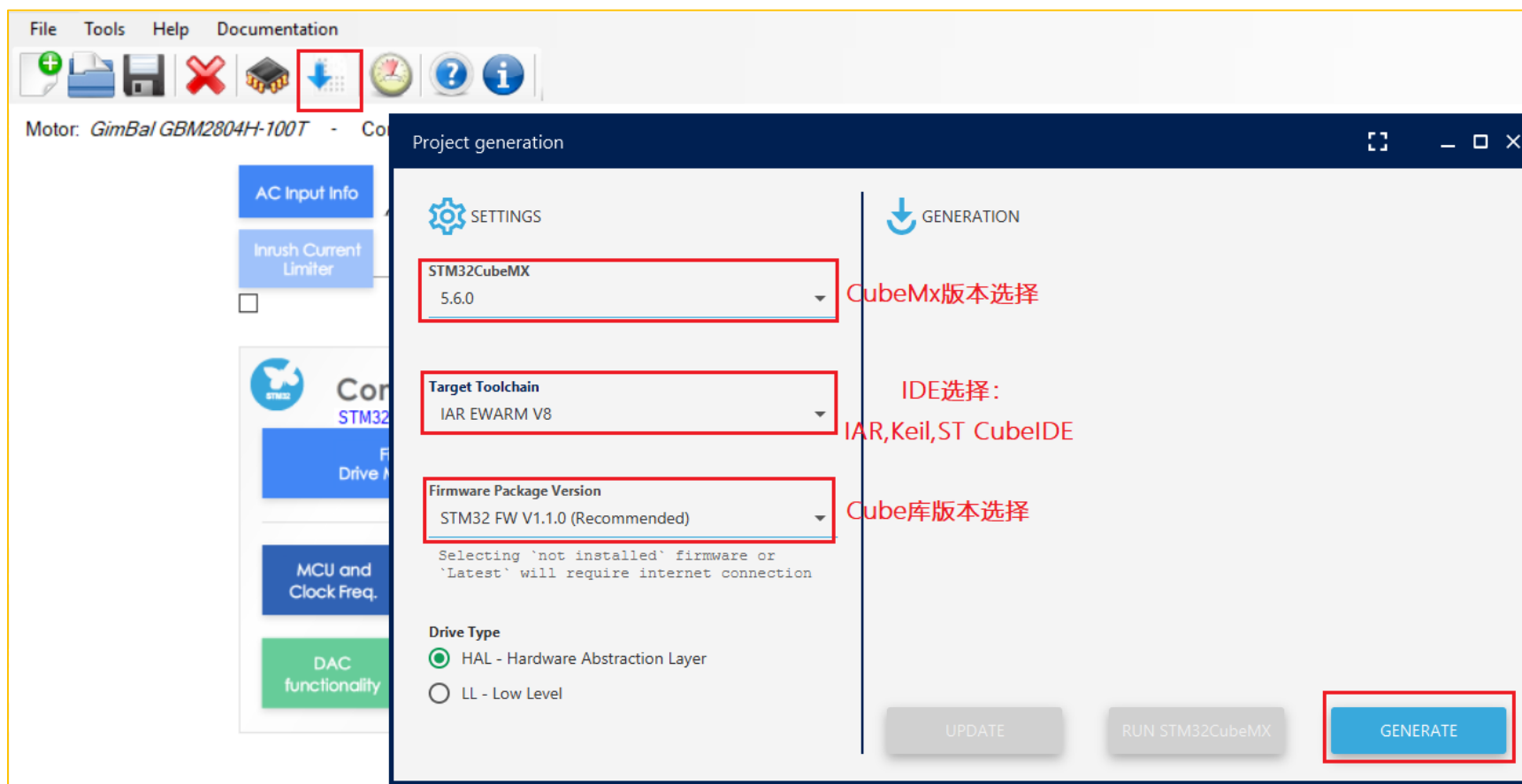


修改相关配置

- 如果使用ST整套Demo板可以**直接生成工程**
- 如果使用的是自己的板子则可以根据电机培训资料上所述进行配置

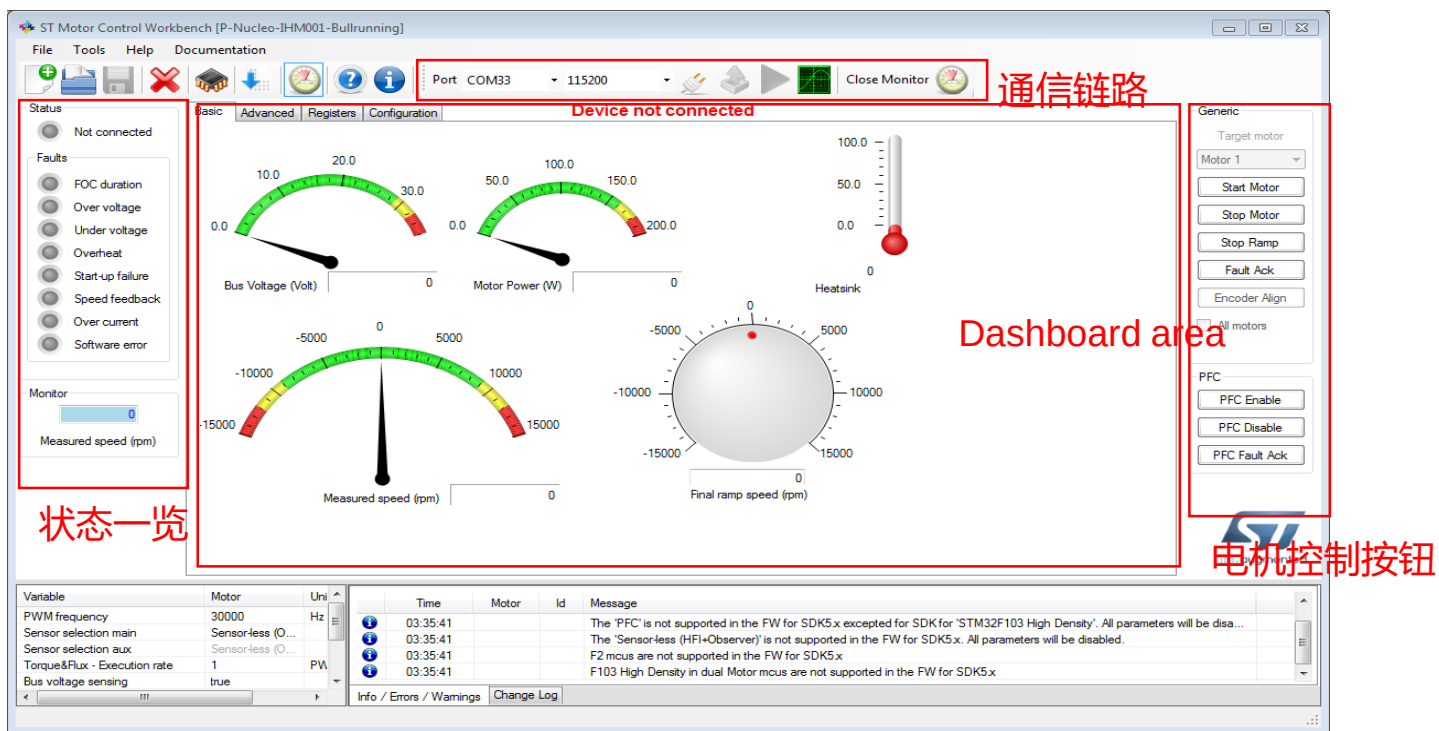


- 选择相应配置生成电机工程



运行电机

- 方法一：使用Workbench串口控制
- 方法二：使用API函数控制(mc_api.c)

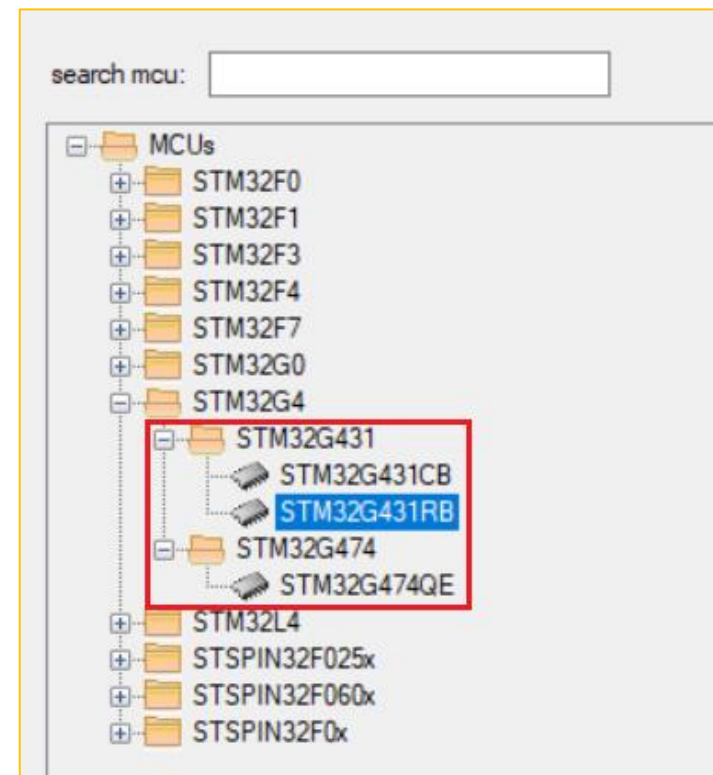


```
mc_api.c
MC_StartMotor1(void)
MC_StopMotor1(void)
MC_ProgramSpeedRampMotor1( int16_t hFinalSpeed, uint16_t hDurationms )
MC_ProgramTorqueRampMotor1( int16_t hFinalTorque, uint16_t hDurationms )
MC_SetCurrentReferenceMotor1( qd_t lqdrref )
MC_GetCommandStateMotor1( void )
MC_StopSpeedRampMotor1(void)
MC_StopRampMotor1(void)
MC_HasRampCompletedMotor1(void)
MC_GetMecSpeedReferenceMotor1(void)
MC_GetMecSpeedAverageMotor1(void)
MC_GetLastRampFinalSpeedMotor1(void)
MC_GetControlModeMotor1(void)
MC_GetImposedDirectionMotor1(void)
MC_GetSpeedSensorReliabilityMotor1(void)
MC_GetPhaseCurrentAmplitudeMotor1(void)
MC_GetPhaseVoltageAmplitudeMotor1(void)
MC_GetlabMotor1(void)
MC_GetIalalphaMotor1(void)
MC_GetIqdMotor1(void)
MC_GetIqdrrefMotor1(void)
MC_GetVqdMotor1(void)
MC_GetValphabetaMotor1(void)
MC_GetElAngledppMotor1(void)
MC_GetTerefMotor1(void)
MC_SetIdrefMotor1( int16_t hNewIdref )
MC_Clear_IqdrrefMotor1(void)
MC_AcknowledgeFaultMotor1( void )
MC_GetOccurredFaultsMotor1(void)
MC_GetCurrentFaultsMotor1(void)
MC_GetSTMStateMotor1(void)
```

基于CubeMx移植电机控制代码

芯片型号未包含在Workbench中的处理

- Workbench中芯片型号选择有限
- 实际运用过程中我们可能用到不同的芯片
- 结合CubeMx和Workbench让我们轻松建立电机工程
- 本例中我们创建STM32G474RET6的工程
 - 控制板：Nucleo-STM32G474RET6
 - 功率板：X-Nucleo-IHM16M1
 - 电机： GBM2804H-100T



MC Workbench操作(1/6)

- 创建Customer Board工程
- 功率板和电机用现有的型号

1 **Application type**
Custom

2 **System**
☒ Single Motor ☐ Dual Motors

3 **Select Boards:** ☐ Inverter ☐ MC Kit ☒ Power & Control

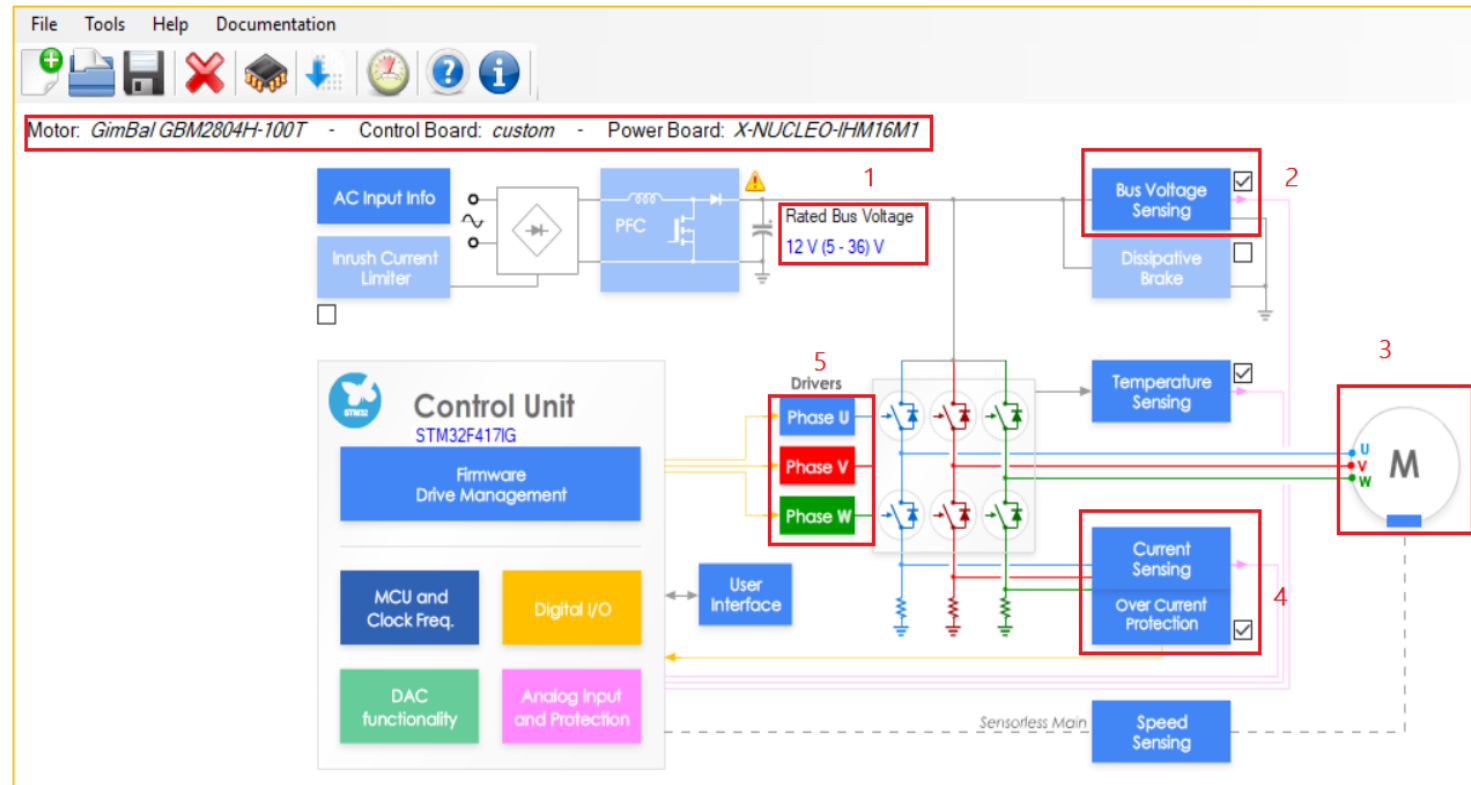
Control
custom board
Control board where the control stage parameters have to be customized by the user

Power
X-NUCLEO-IHM16M1 3Sh
based on STSPIN830
DC Input voltage 7 - 45 Vdc
Output pk current 0.1 - 2.12 Apk
Nominal Power 1 - 60 W
Active

4 **Motor**
Gimbal GBM2804H-100T
iPower GBM2804H-100T Brushless Gimbal Motor
Magnetic structure Surface Mounted
Pole Pairs 7
Nominal Speed 1572 rpm
Nominal Voltage 12.1 V
Nominal Current 0.15 Apk

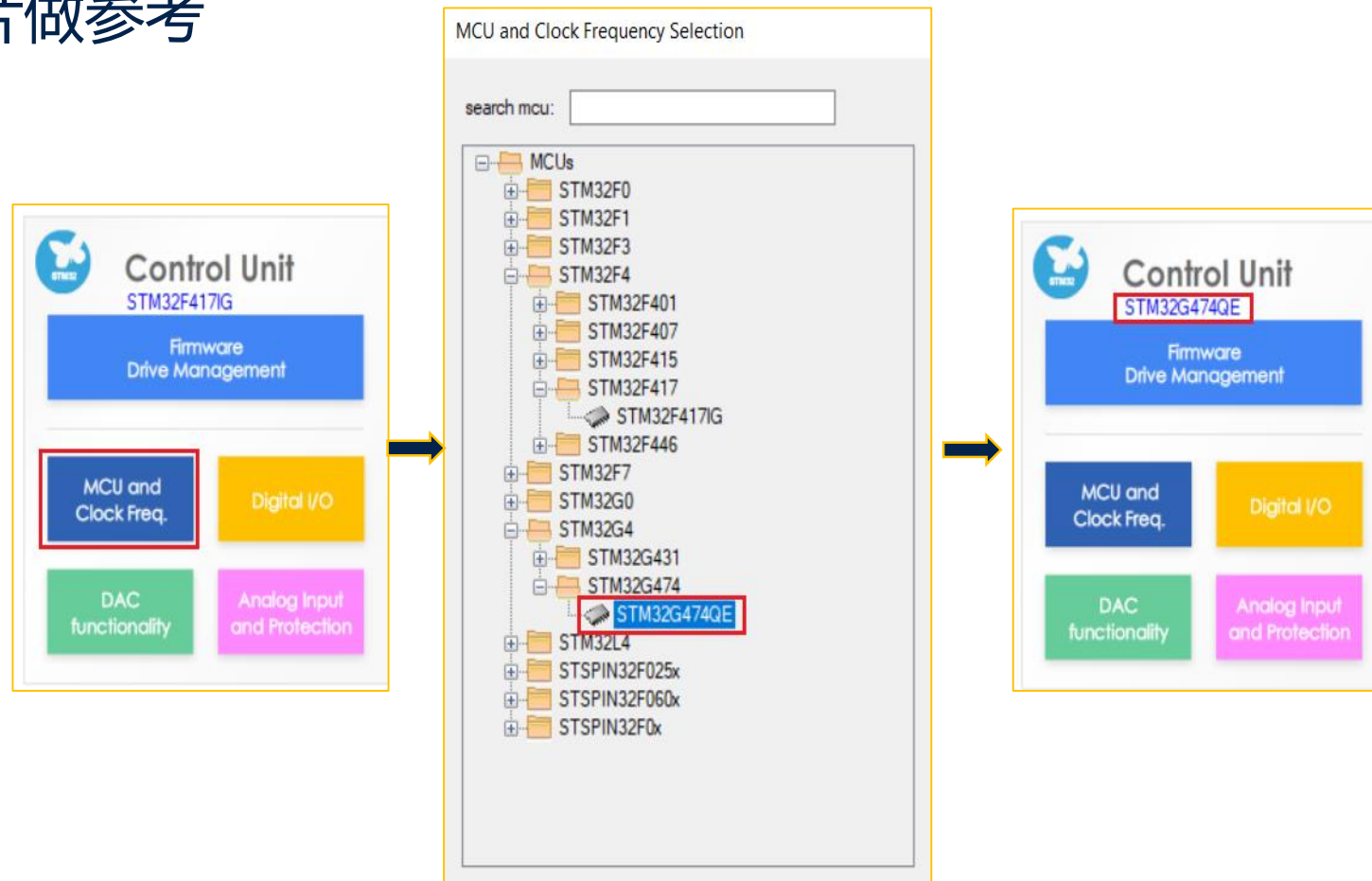
MC Workbench操作(2/6)

- 配置电机以及功率板相关参数
 - 这边我们使用默认的配置，如果是自己的板子，需要重新配置



MC Workbench操作(3/6)

- 芯片选择与配置
- 找到类似芯片做参考



MC Workbench操作(4/6)

• 数字端口配置

- 主要是Timer管脚，控制脚，串口等
- 有传感的控制还需要配置编码器和Hall信号

Control Stage - Digital I/O

Control Unit
STM32G474QE

Firmware Drive Management

MCU and Clock Freq.

Digital I/O

DAC functionality

Analog Input and Protection

Inverter driving signal selection

Timer: TIM1

Remap: No remap

Pin Map

Channel	Pin
CH1	A8
CH2	A9
CH3	A10
BKIN2	A11

CH1N: E8

CH2N: E10

CH3N: E12

Signal Enabler

Channel	Port	Pin
CH1	GPIOB	B13
CH2	GPIOB	B14
CH3	GPIOB	B15

Speed/position feedback

Encoder interface

Timer: TIM2

Remap: No remap

Pin Map

Channel	Pin
CH1	A0
CH2	B3

Hall sensors interface

Timer: TIM2

Remap: No remap

Pin Map

Channel	Pin
CH1	A0
CH2	B3
CH3	B10

Direct GPIO

DBO

Port: GPIOE

Pin: E5

ICL

Port: GPIOE

Pin: E4

OCP disabling

Port: GPIOE

Pin: E5

PFC drive signal and feedback

Timer:

Pin Map

Signal	Pin
PWM	--
AC Mains	--
OCS	--

Serial communication

Channel: USART2

Baudrate: 115200

Remap: No remap

Pin Map

Signal	Pin
TX	A2
RX	A3

Start/Stop Button GPIO

Port: GPIOC

Pin: C13

Trigger edge: Falling Edge

Internal Pull up/down: No Pull up - No Pull down

Done

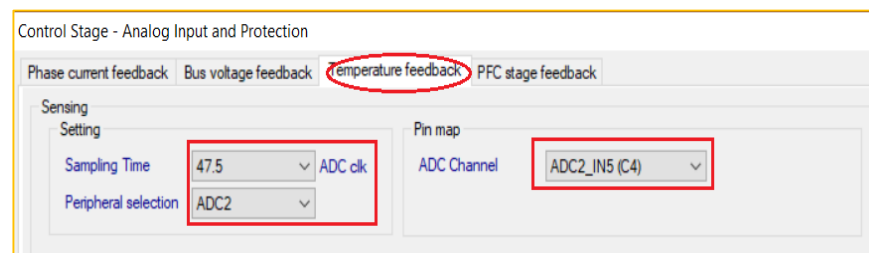
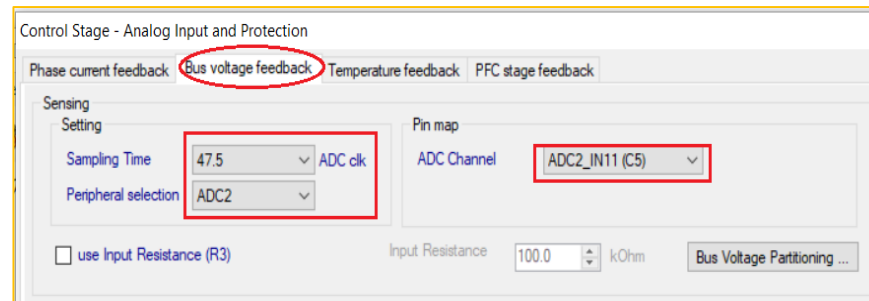
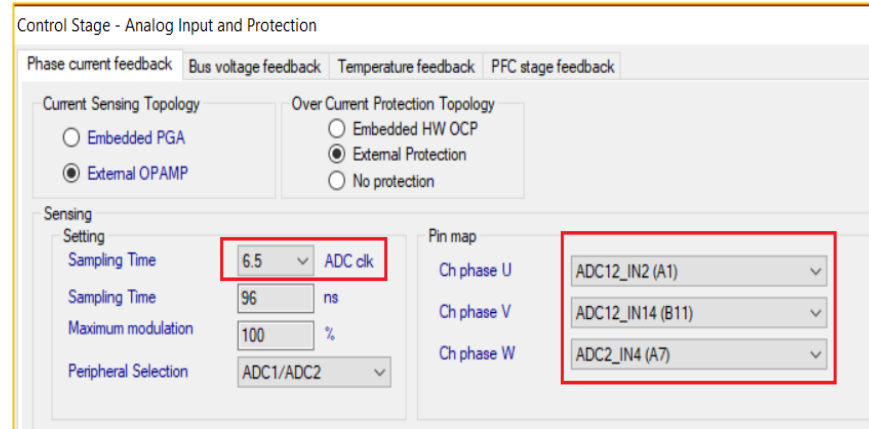
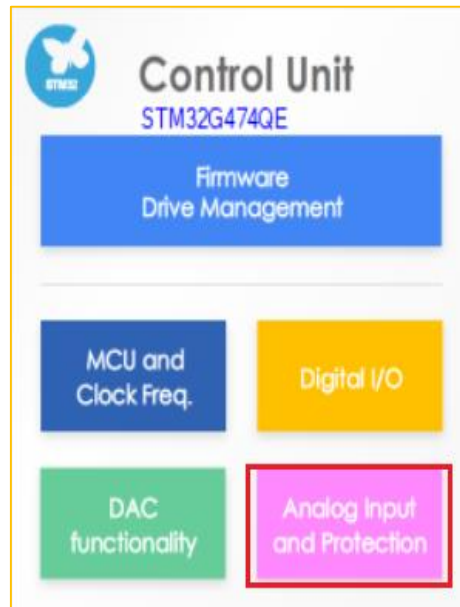
管脚控制使能管脚选择

CHxN--PWM互补模式才可选择

调试串口配置

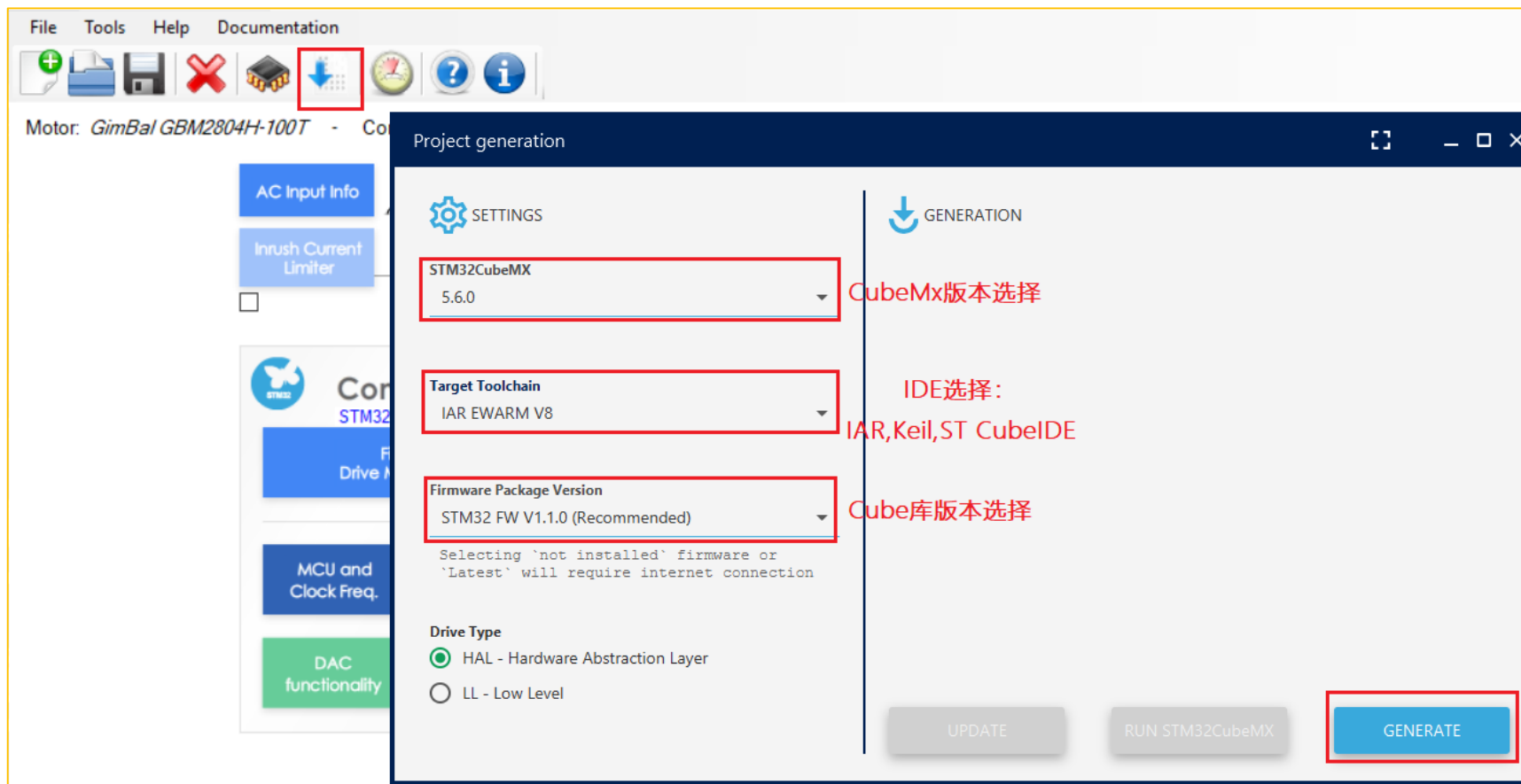
MC Workbench操作(5/6)

- 模拟端口配置
 - 电流, 电压, 温度



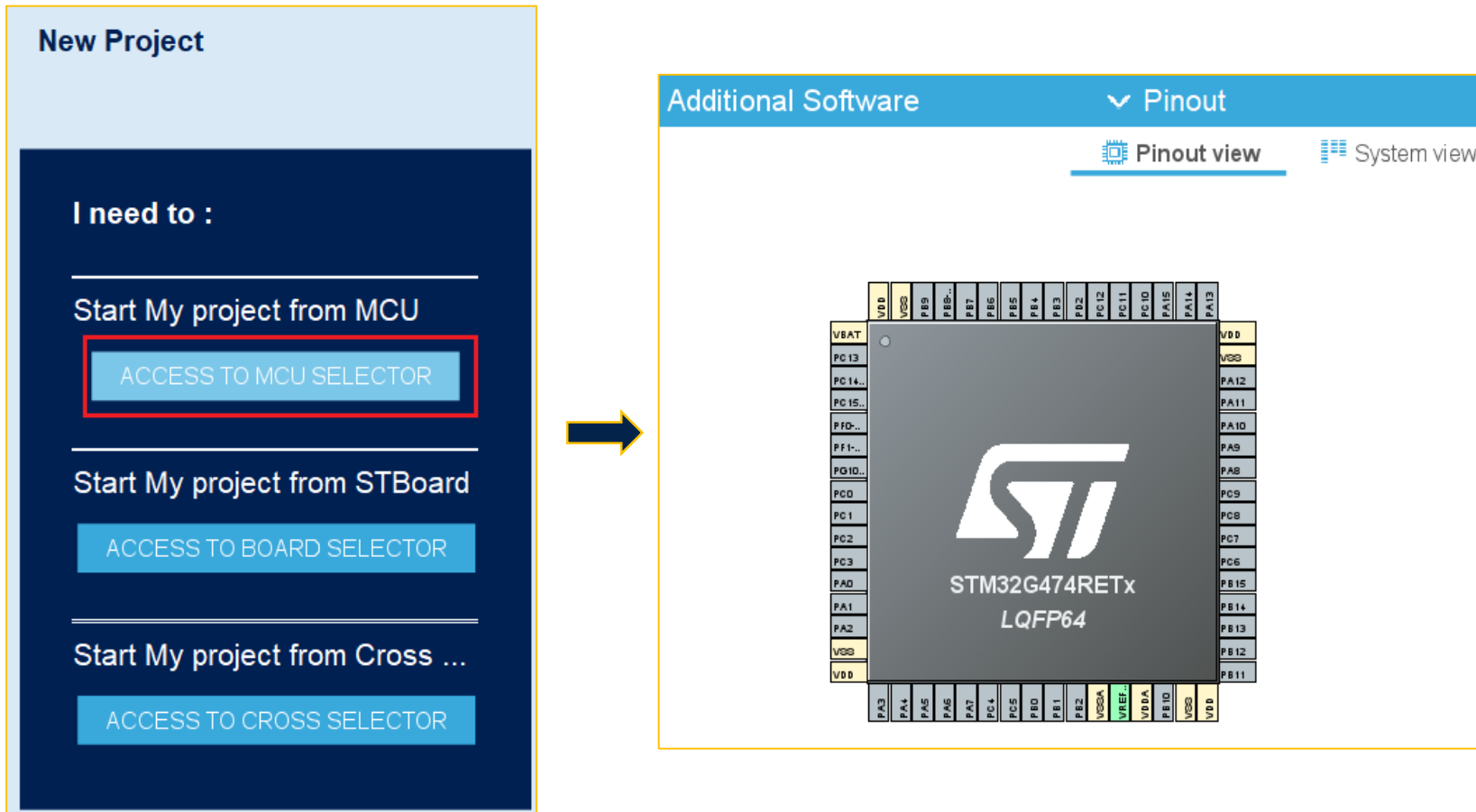
MC Workbench操作(6/6)

- 生成参考代码



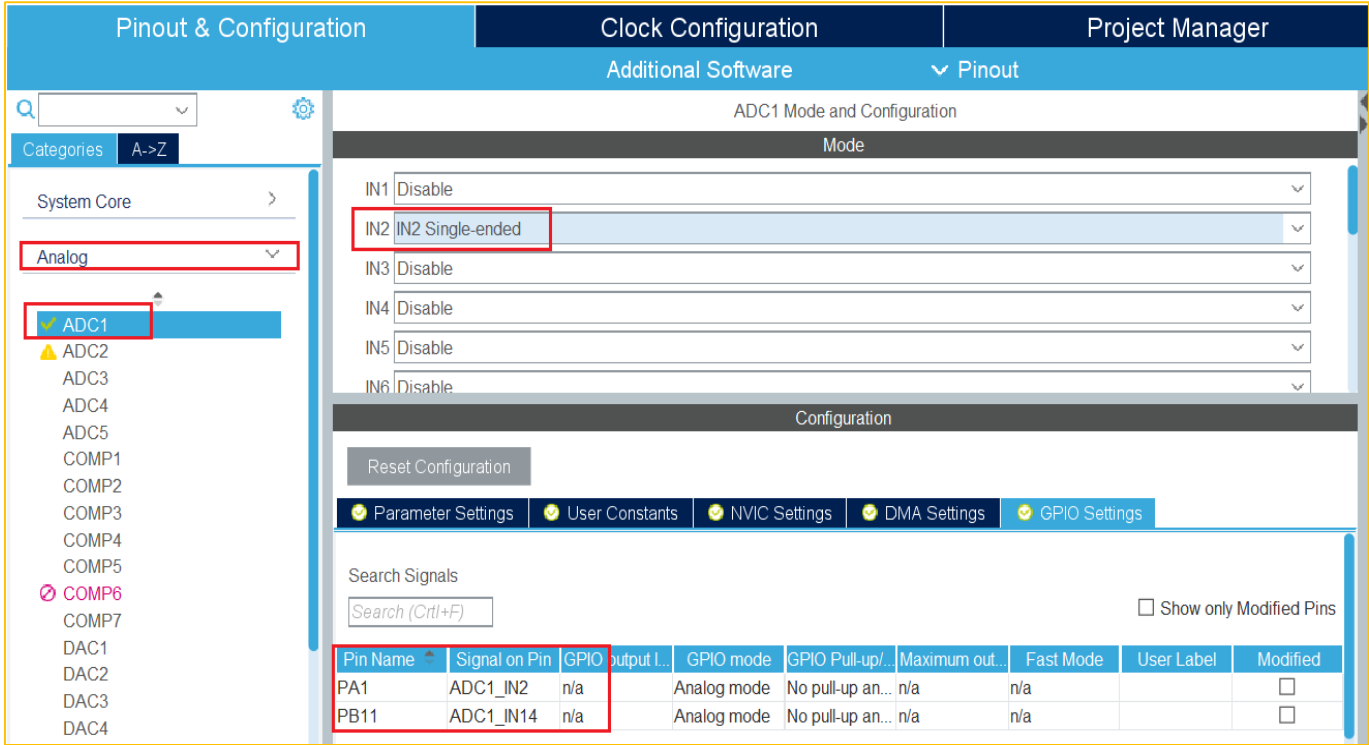
CubeMx操作(1/16)

- 新建STM32G474RET6的CubeMx工程



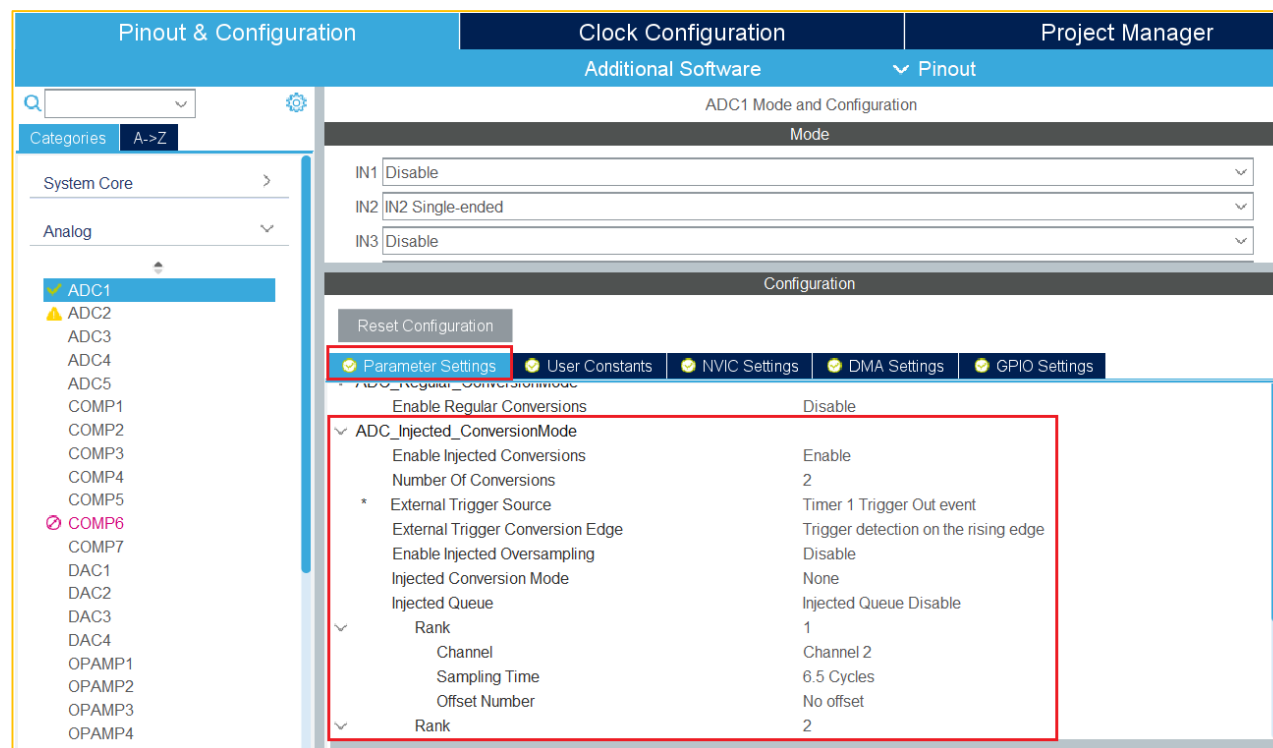
CubeMx操作(2/16)

- 配置STM32G474RET6的ADC管脚
 - PA1--ADC1_IN2 (CU) , PB11--ADC1_IN14 (CV)
- 可以打开上面Workbench中生成的电机工程参考配置



- ADC1功能配置

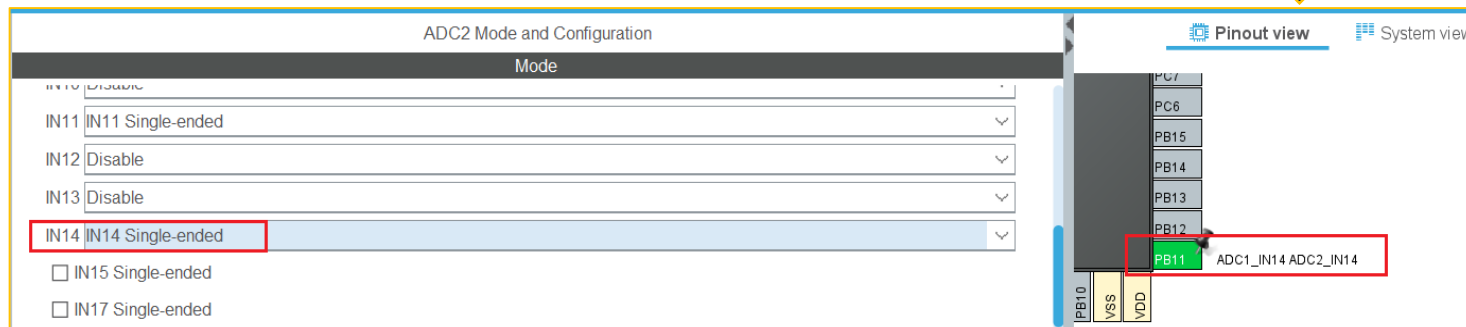
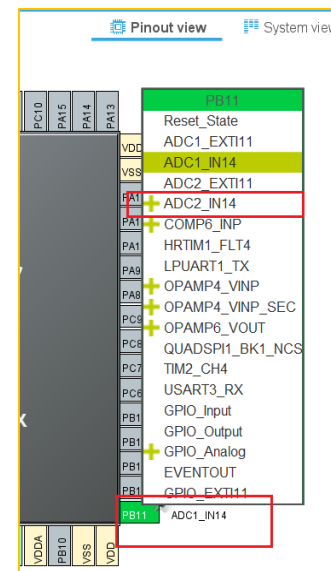
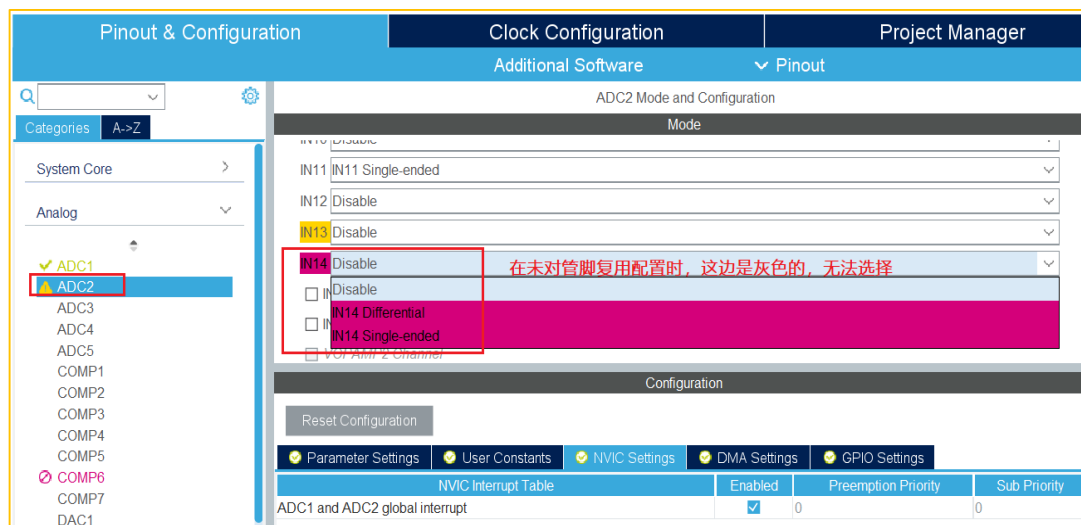
- 左对齐, 触发信号为Tim1 Trigger Out Event
- ADC1只有 Injected模式没有Regular模式
- Rank1:Channel2 ; Rank2:Channel 14



CubeMx操作(4/16)

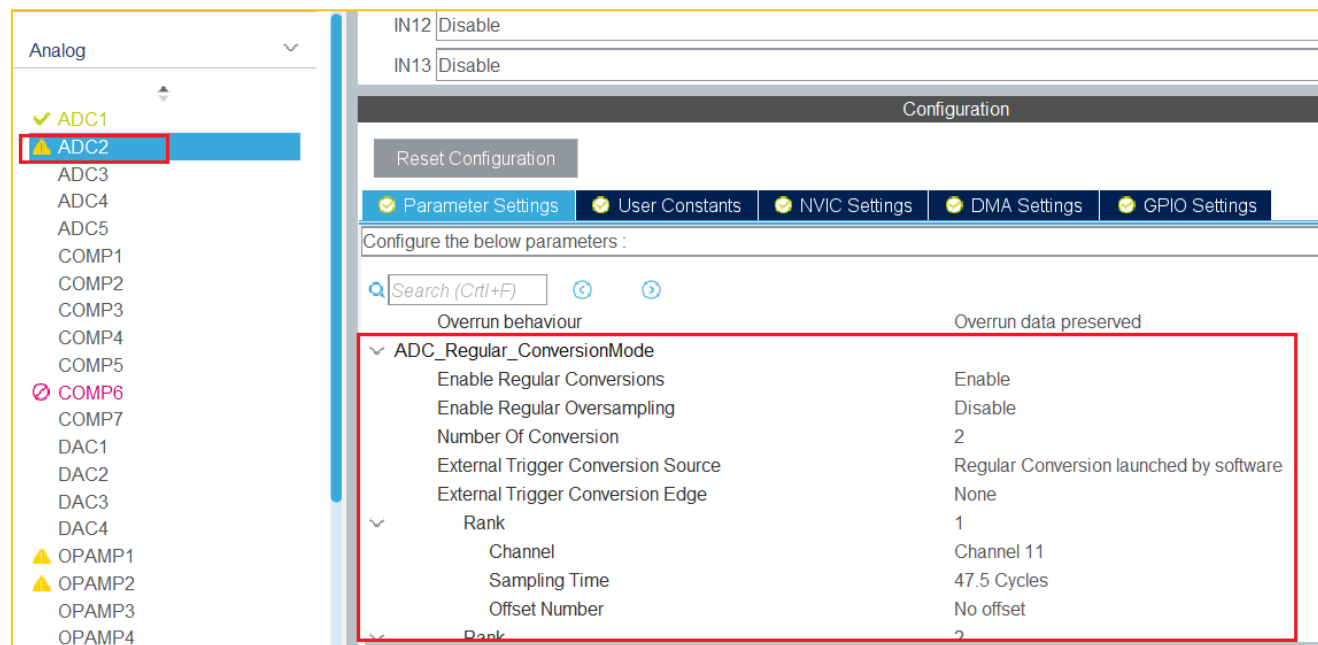
配置ADC2引脚

- 注入组: PA7--ADC2_IN4 (CW)
PB11--ADC2_IN14 (CV)
- 规则组: PC5--ADC2_IN11 (Bus)
PC4--ADC2_IN5 (Temp)
- 这边ADC2_IN14复用ADC1_IN14的引脚



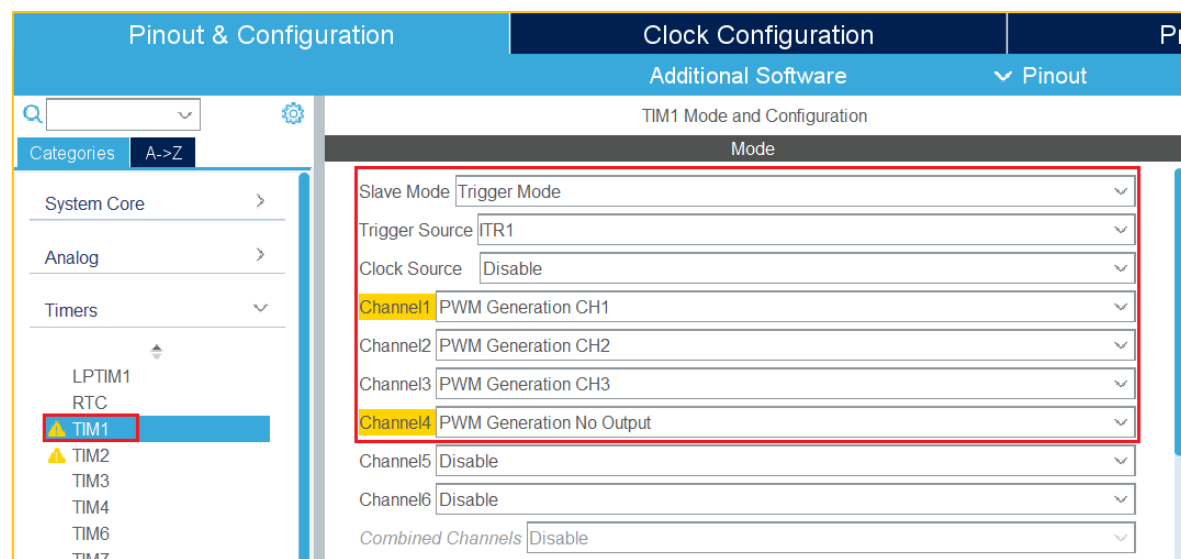
- 配置ADC2功能

- ADC2有两个通道用于电压与温度传感采样，使用Regular模式
- 电流采样的使用 Injected模式



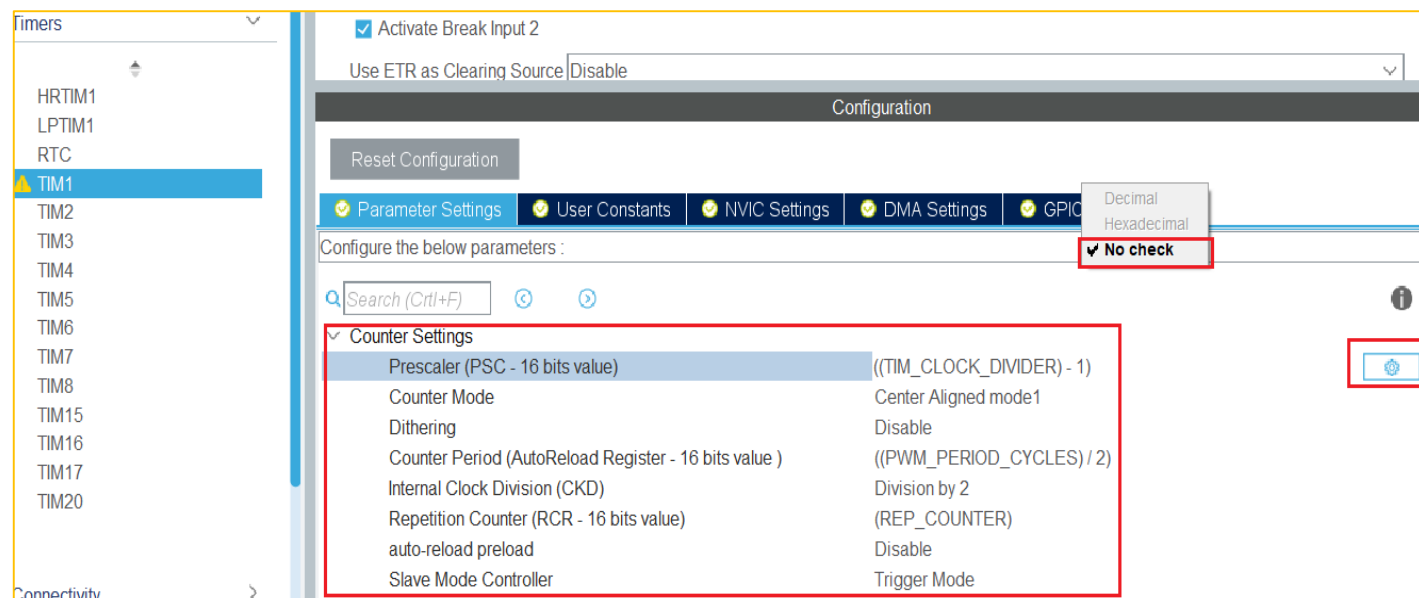
CubeMx操作(6/16)

- 配置TIM1输出管脚
 - 三路上桥输出PWM波
 - 其中CH4用于ADC触发



- 配置TIM1功能

- 这边宏定义，可以参考MC Workbench生成的工程
- 注意定义为相同名字



- 使能ADC1/2中断

Parameter Settings	User Constants	NVIC Settings	DMA Settings	GPIO Settings
NVIC Interrupt Table		Enabled	Preemption Priority	
ADC1 and ADC2 global interrupt		☒	0	

- 使能TIM1中断

Reset Configuration

Parameter Settings

User Constants

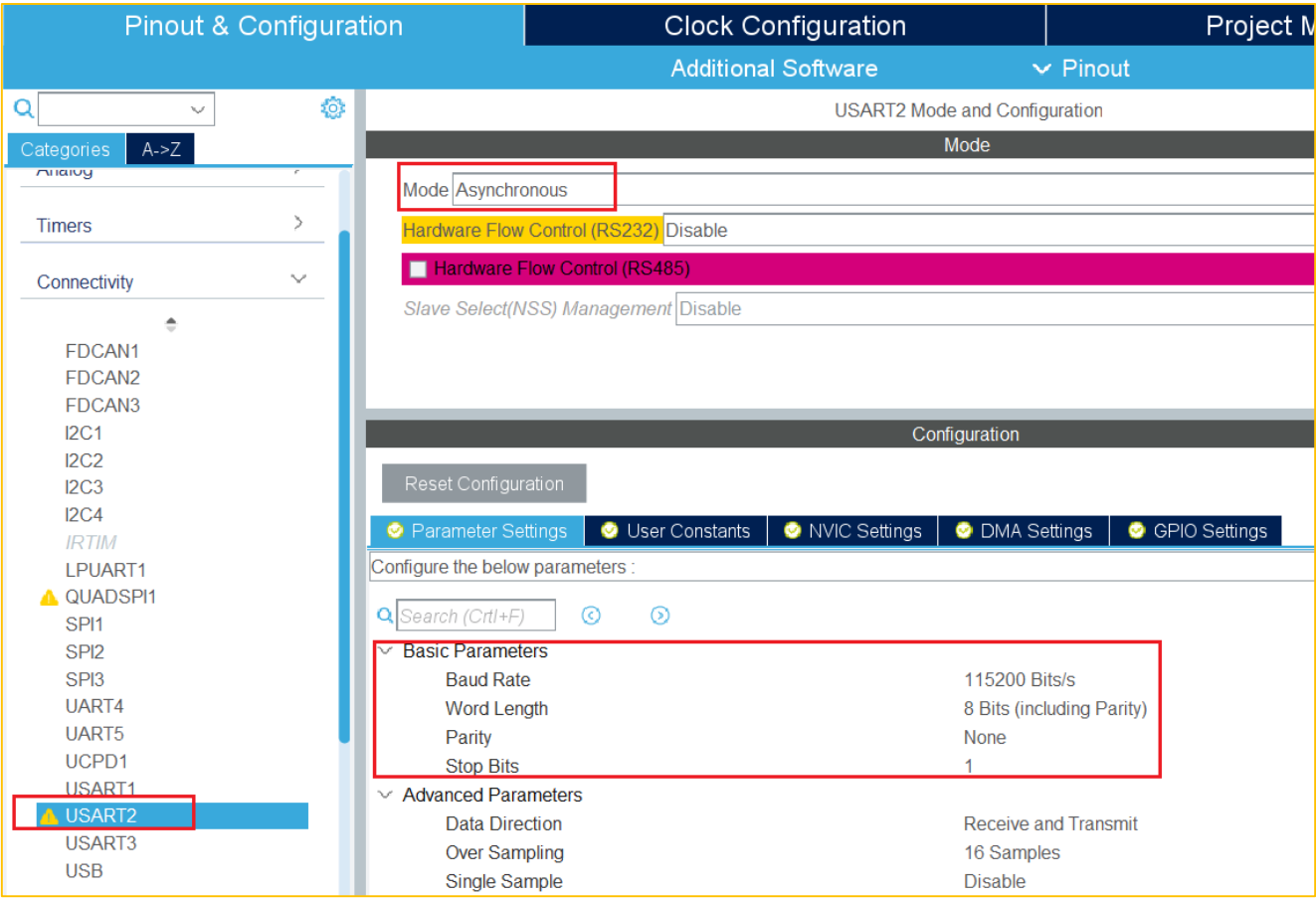
NVIC Settings

DMA Settings

GPIO Settings

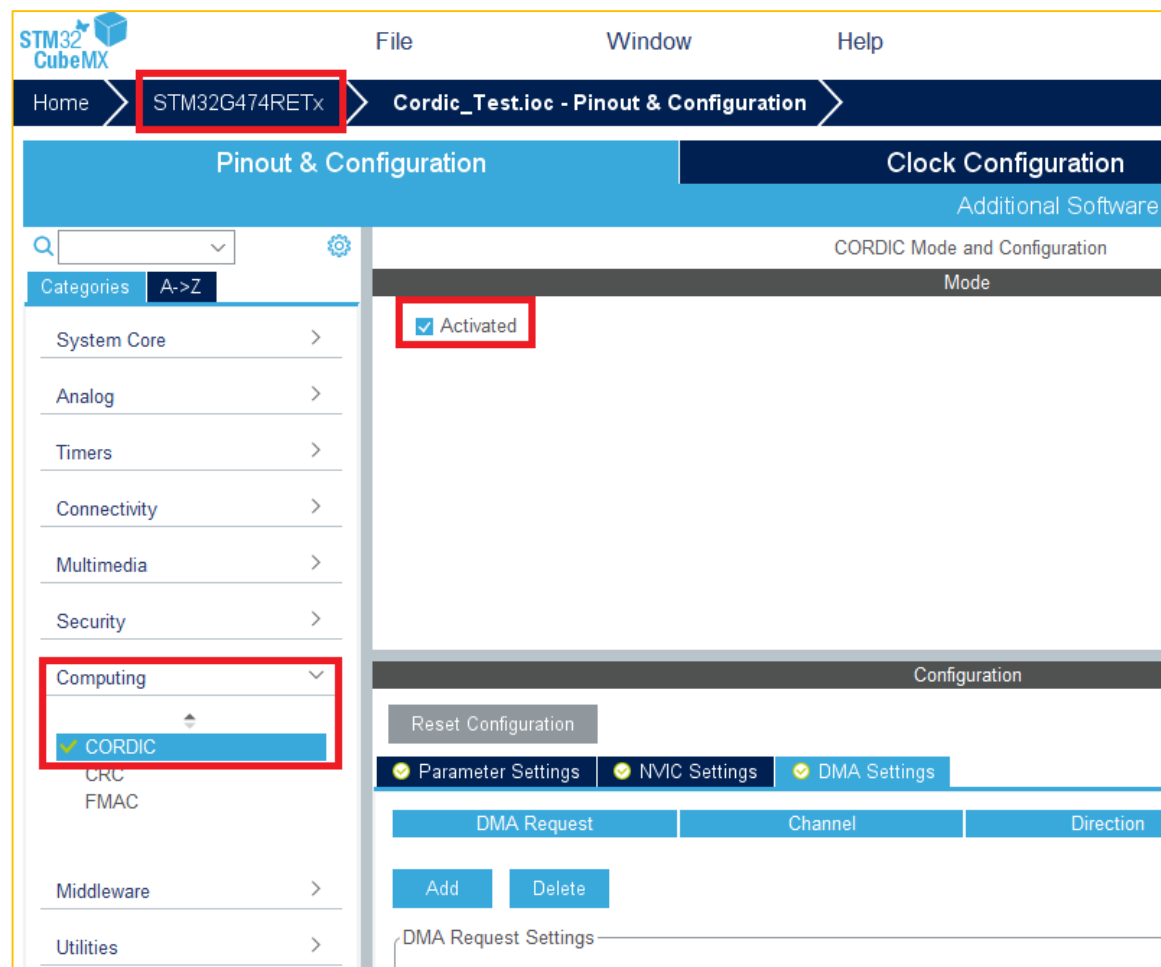
NVIC Interrupt Table	Enabled	Preemption Priority	Sub
TIM1 break interrupt and TIM15 global interrupt	☒	0	0
TIM1 update interrupt and TIM16 global interrupt	☒	0	0
TIM1 trigger and commutation interrupts and TIM17 global interrupt	☐	0	0
TIM1 capture compare interrupt	☐	0	0

- 配置调试串口



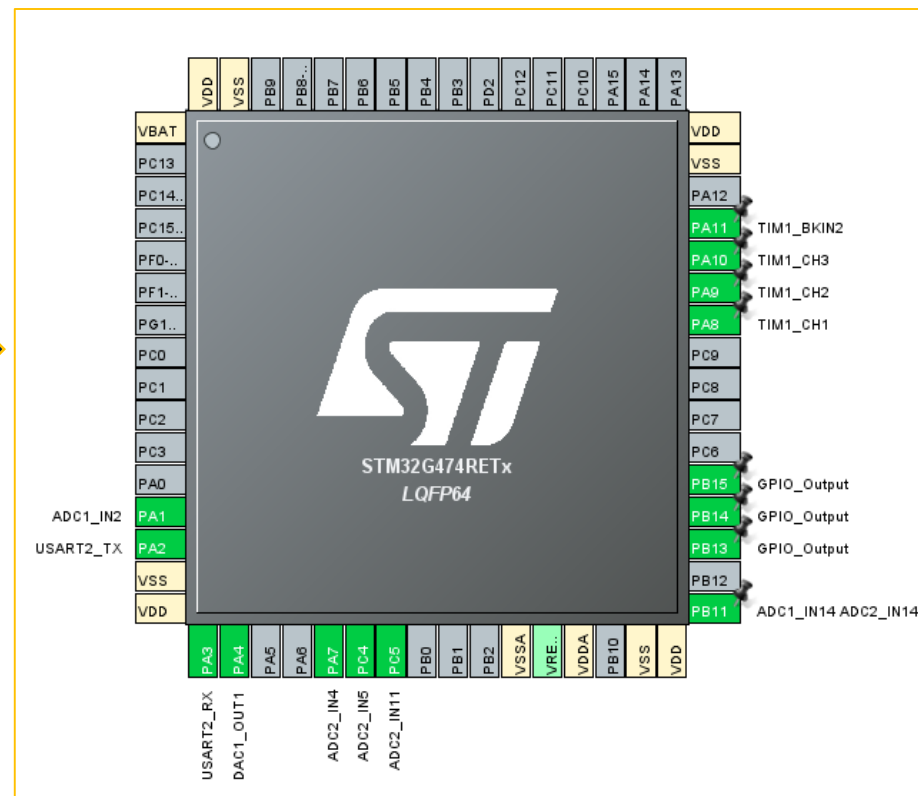
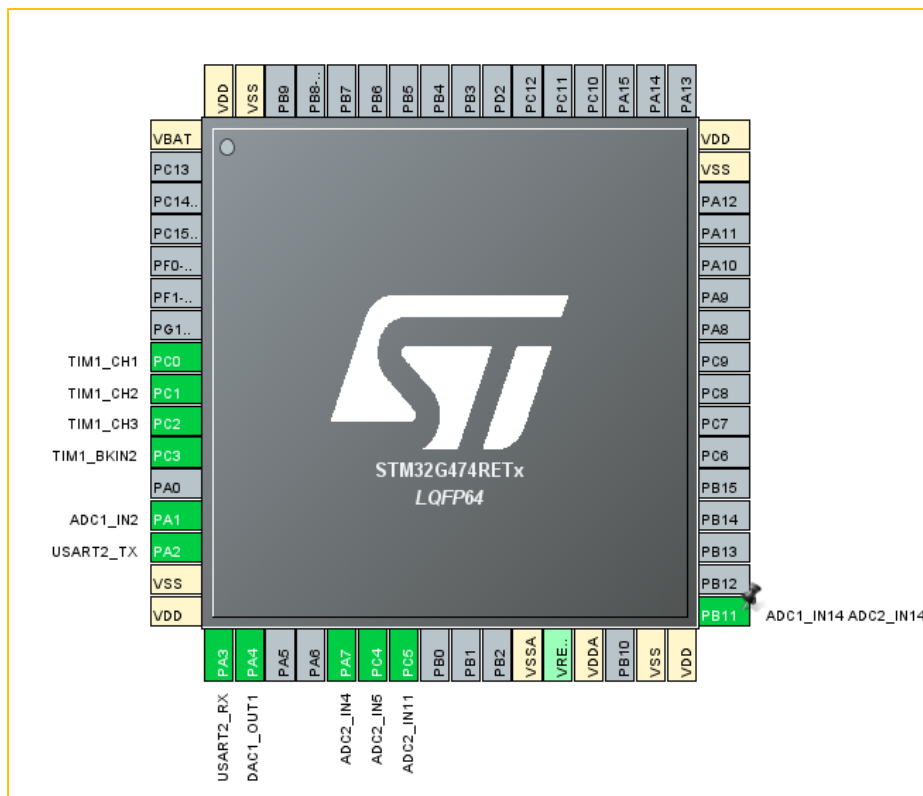
CubeMx操作(10/16)

- 使能STM32G4特有的Cordic外设



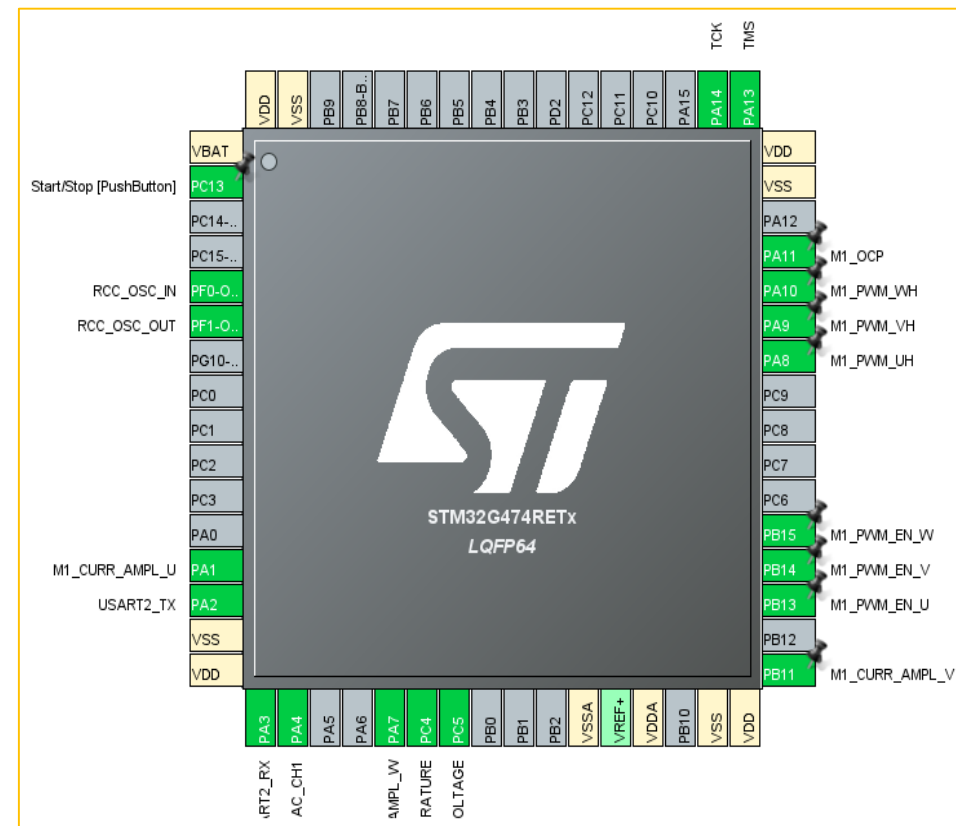
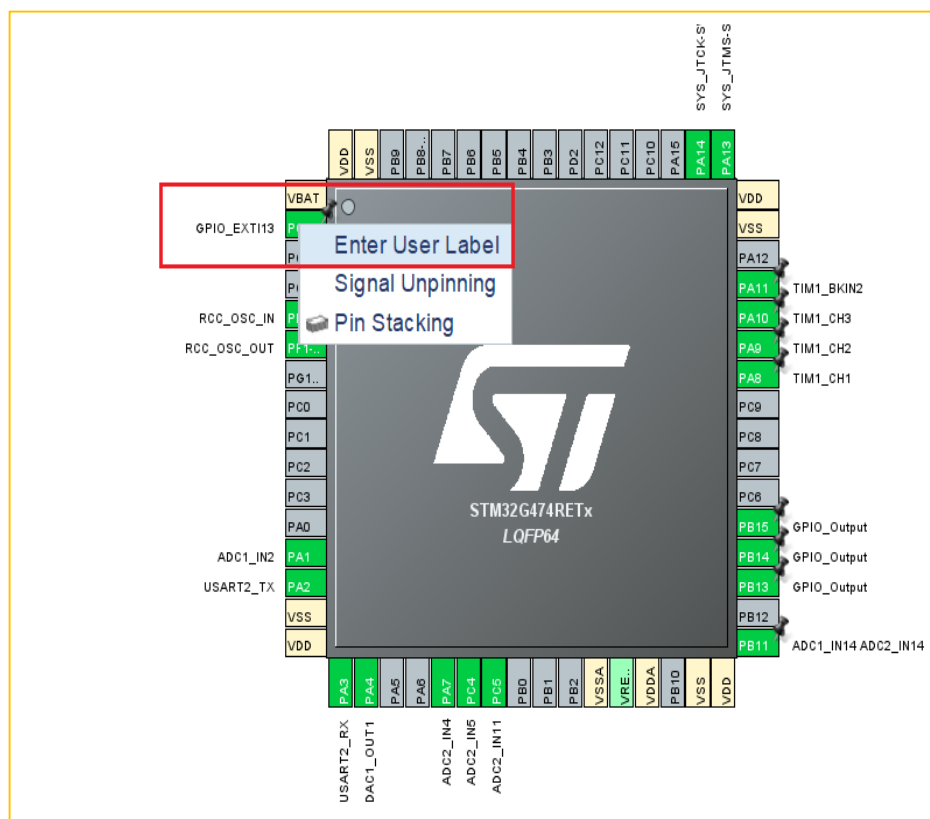
CubeMx操作(11/16)

- 重新分配管脚
- 根据实际电路进行配置

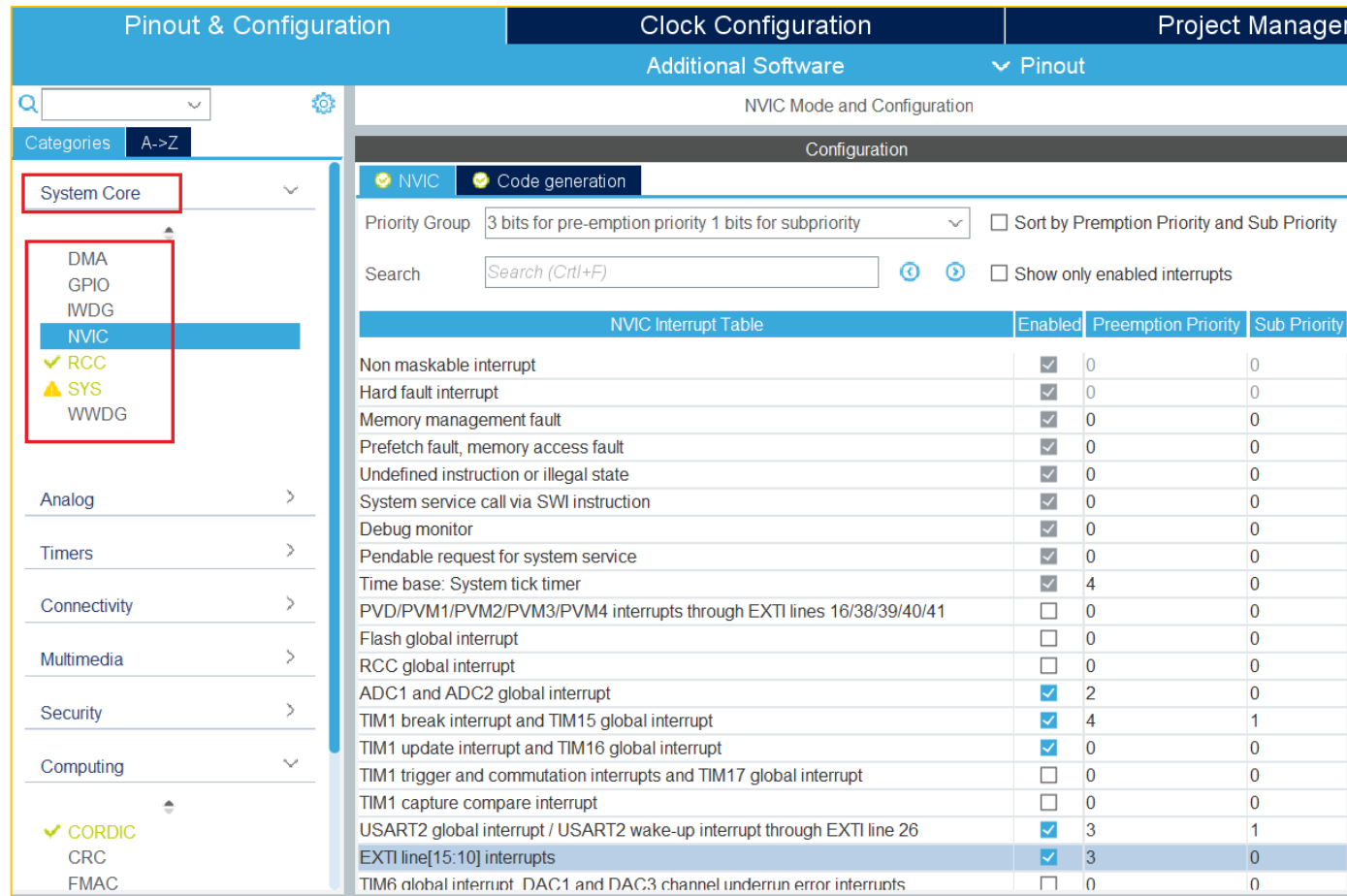


CubeMx操作(12/16)

- 管脚名称重定义
- 为方便使用，定义为MC Workbench生成的命名



- 规划NVIC的中断优先级



The screenshot displays the STM32CubeMX software interface for configuring the NVIC (Nested Vectored Interrupt Controller). The left sidebar shows the 'System Core' category with 'NVIC' selected. The main panel shows the 'NVIC Mode and Configuration' tab with a table of interrupts and their priorities.

NVIC Mode and Configuration

Configuration

Priority Group: 3 bits for pre-emption priority 1 bits for subpriority

Search: Search (Ctrl+F)

☐ Sort by Preemption Priority and Sub Priority

☐ Show only enabled interrupts

NVIC Interrupt Table	Enabled	Preemption Priority	Sub Priority
Non maskable interrupt	☒	0	0
Hard fault interrupt	☒	0	0
Memory management fault	☒	0	0
Prefetch fault, memory access fault	☒	0	0
Undefined instruction or illegal state	☒	0	0
System service call via SWI instruction	☒	0	0
Debug monitor	☒	0	0
Pendable request for system service	☒	0	0
Time base: System tick timer	☒	4	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	☐	0	0
Flash global interrupt	☐	0	0
RCC global interrupt	☐	0	0
ADC1 and ADC2 global interrupt	☒	2	0
TIM1 break interrupt and TIM15 global interrupt	☒	4	1
TIM1 update interrupt and TIM16 global interrupt	☒	0	0
TIM1 trigger and commutation interrupts and TIM17 global interrupt	☐	0	0
TIM1 capture compare interrupt	☐	0	0
USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26	☒	3	1
EXTI line[15:10] interrupts	☒	3	0
TIM6 global interrupt DAC1 and DAC3 channel underrun error interrupts	☐	0	0

- 不生成电机控制相关的中断程序

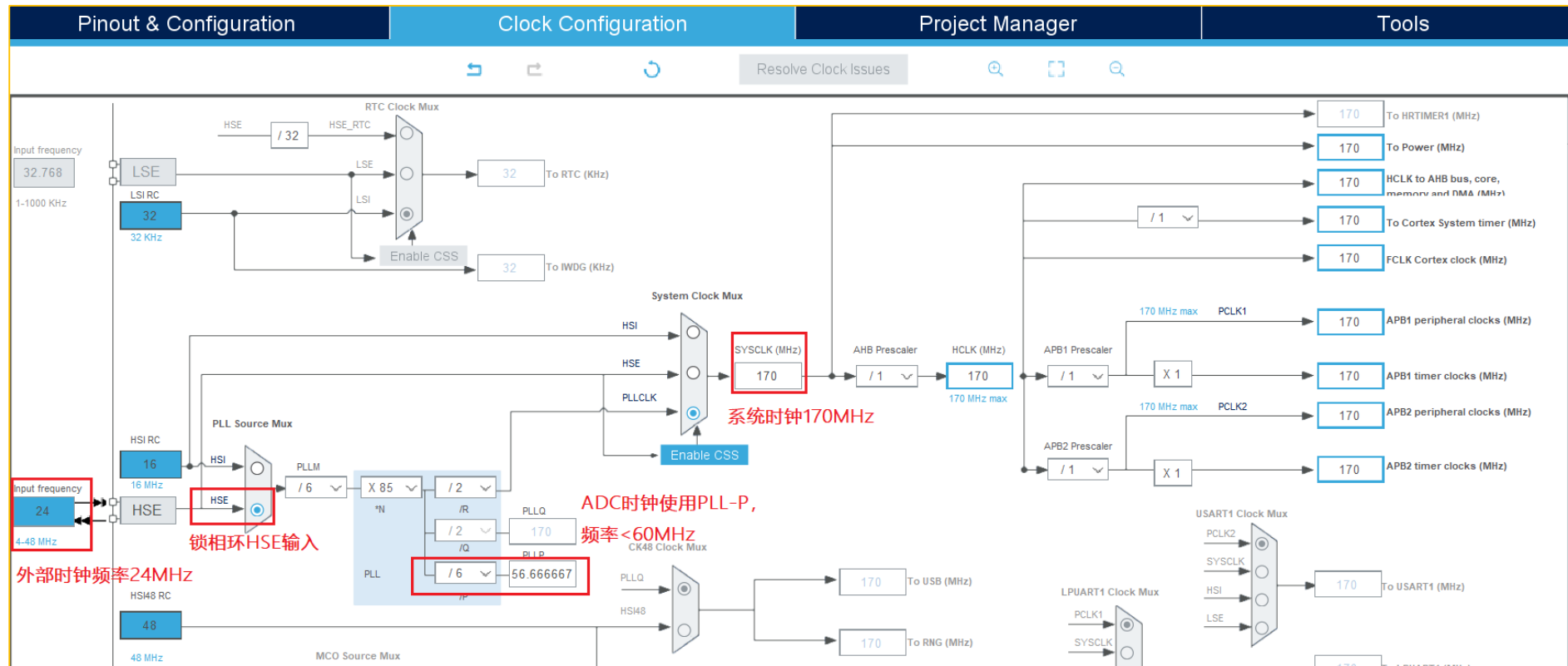
Configuration			
☒ NVIC	☒ Code generation		
Enabled interrupt table	☒ Select for init sequence ordering	☐ Generate IRQ handler	Call HAL handler
Debug monitor	☐	☐	☐
Pendable request for system service	☐	☐	☐
Time base: System tick timer	☐	☐	☒
ADC1 and ADC2 global interrupt	☒	☐	☒
TIM1 break interrupt and TIM15 global interrupt	☒	☐	☒
TIM1 update interrupt and TIM16 global interrupt	☒	☐	☒
USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26	☒	☐	☒
EXTI line[15:10] interrupts	☒	☐	☒

Interrupt unmasking ordering table (interrupt init code is moved after all the peripheral init code)

CubeMx操作(15/16)

• 时钟配置

- Demo板上为24MHz外部时钟
- 系统时钟配置为170MHz



CubeMx操作(16/16)

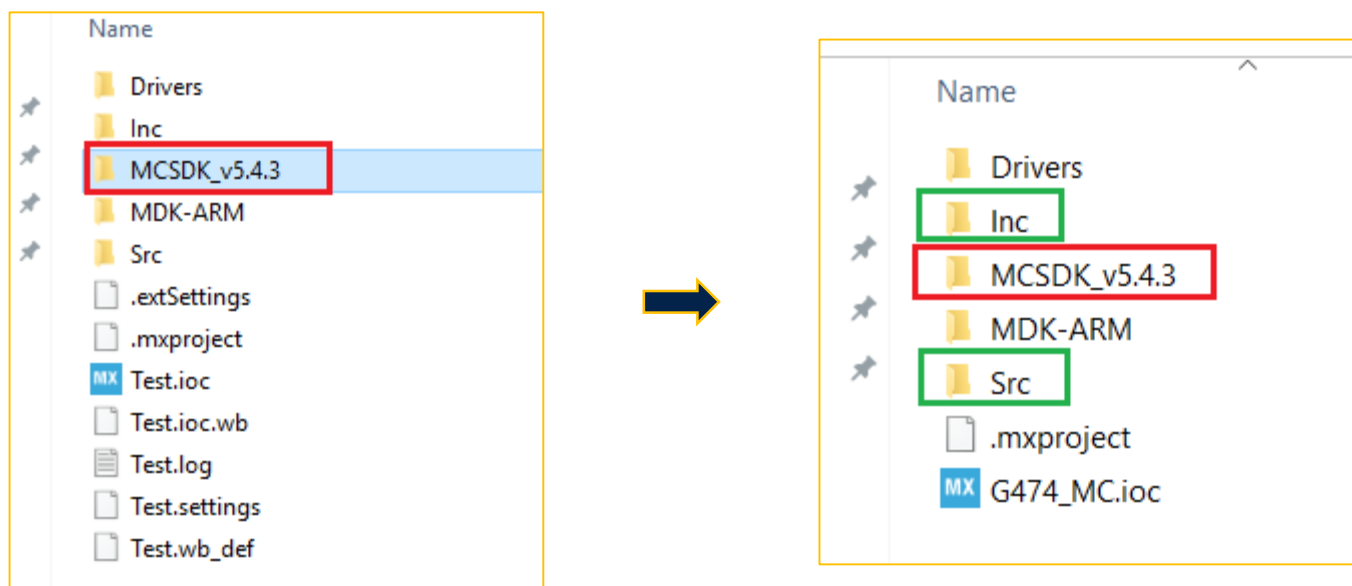
- CubeMx生成工程

The screenshot displays the STM32CubeMX Project Manager interface for a project named 'Cube_G4'. The breadcrumb navigation at the top shows 'Home' > 'STM32G474RETx' > 'Cube_G4.ioc - Project Manager'. A 'GENERATE CODE' button is located in the top right corner. The main navigation bar includes 'Pinout & Configuration', 'Clock Configuration', 'Project Manager' (which is highlighted with a red box), and 'Tools'. A red handwritten label '生成代码' (Generate Code) is placed over the 'Tools' tab. On the left side, there is a vertical menu with three sections: 'Project', 'Code Generator', and 'Advanced Settings'. The 'Project Settings' section is expanded, showing the following configuration options:

- Project Name:** Cube_G4
- Project Location:** (empty text field)
- Application Structure:** Basic (dropdown menu) with a checkbox for 'Do not generate the main()'.
- Toolchain Folder Location:** (empty text field)
- Toolchain / IDE:** MDK-ARM (dropdown menu)
- Min Version:** V5 (dropdown menu)
- Generate Under Root:** (unchecked checkbox)
- Linker Settings:**
 - Minimum Heap Size:** 0x200
 - Minimum Stack Size:** 0x400

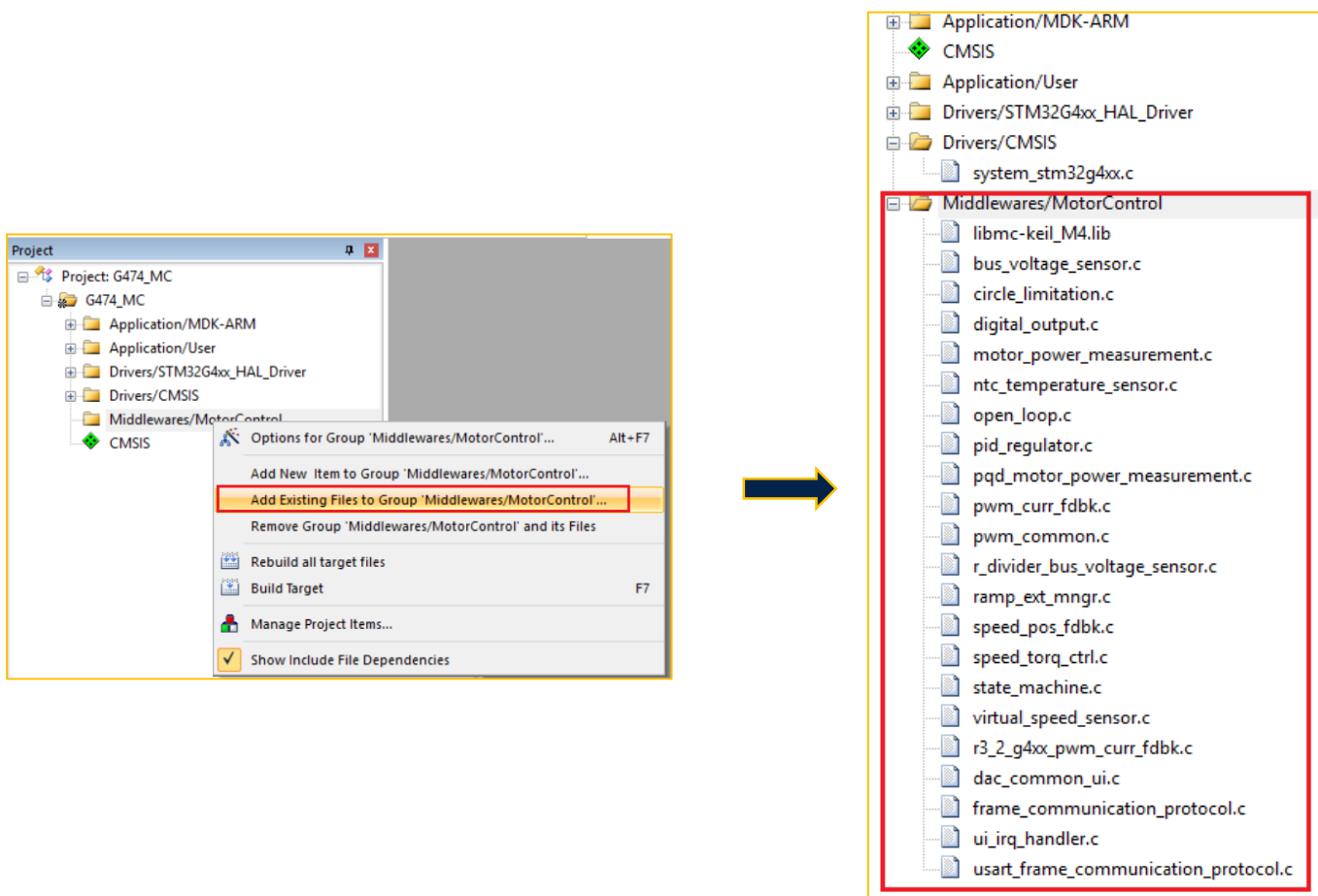
工程修改添加(1/5)

- MCSDK_v5.4.3拷贝到CubeMx直接生成的文件夹下
- Inc与Src文件也同时拷贝进去，需要选择不覆盖同名文件



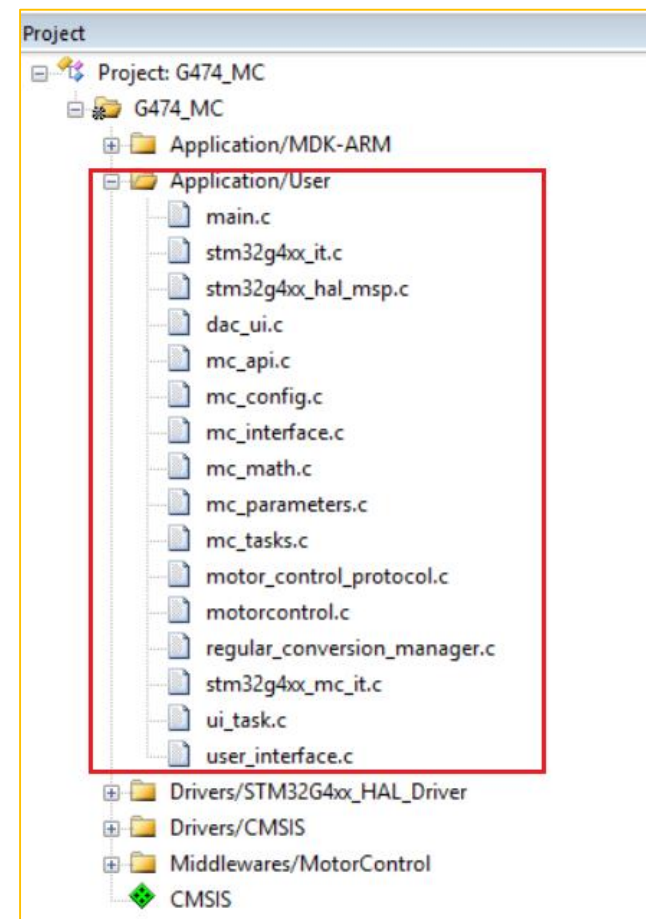
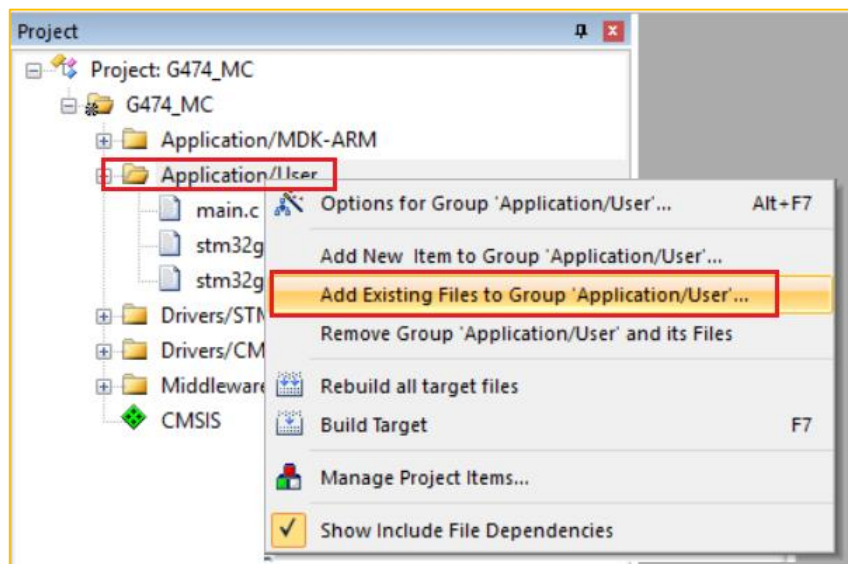
工程修改添加(2/5)

- 对CubeMx生成的工程添加FOC库文件



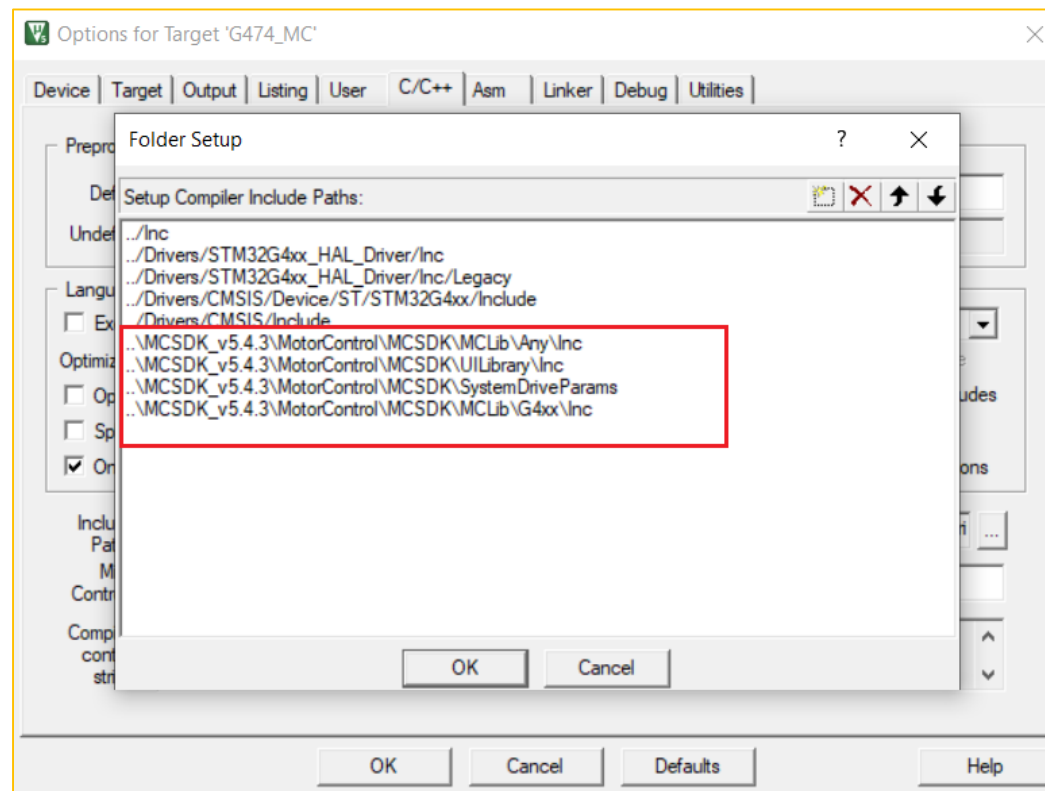
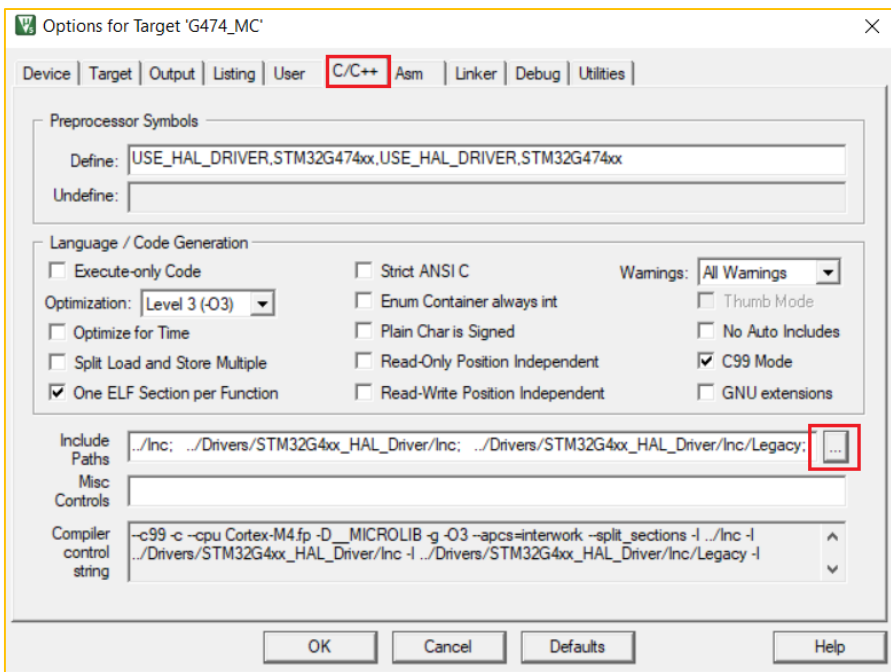
工程修改添加(3/5)

- 增加API以及控制文件到工程中



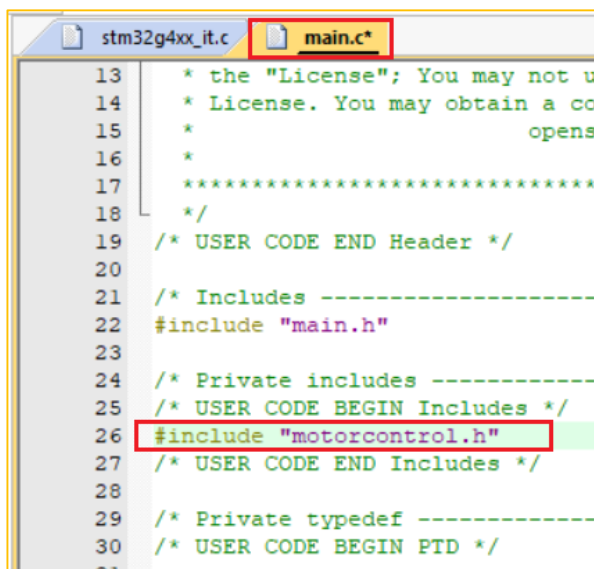
工程修改添加(4/5)

- 增加头文件包含目录

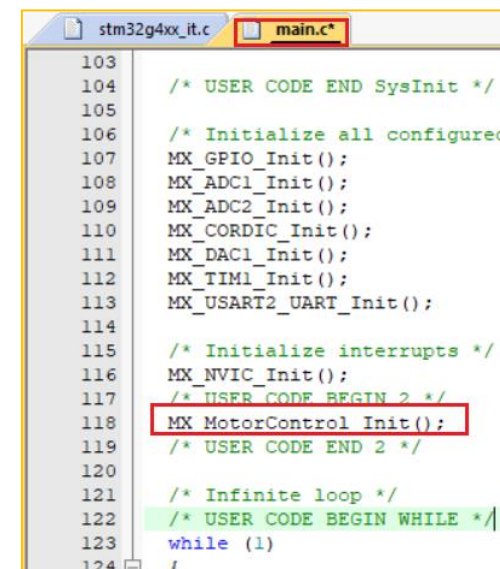


工程修改添加(5/5)

- 修改Main.c文件
 - 包含头文件
 - 增加电机初始化函数



```
13  * the "License"; You may not u
14  * License. You may obtain a co
15  *
16  *
17  *
18  */
19  /* USER CODE END Header */
20
21  /* Includes -----
22  #include "main.h"
23
24  /* Private includes -----
25  /* USER CODE BEGIN Includes */
26  #include "motorcontrol.h"
27  /* USER CODE END Includes */
28
29  /* Private typedef -----
30  /* USER CODE BEGIN PTD */
31
```

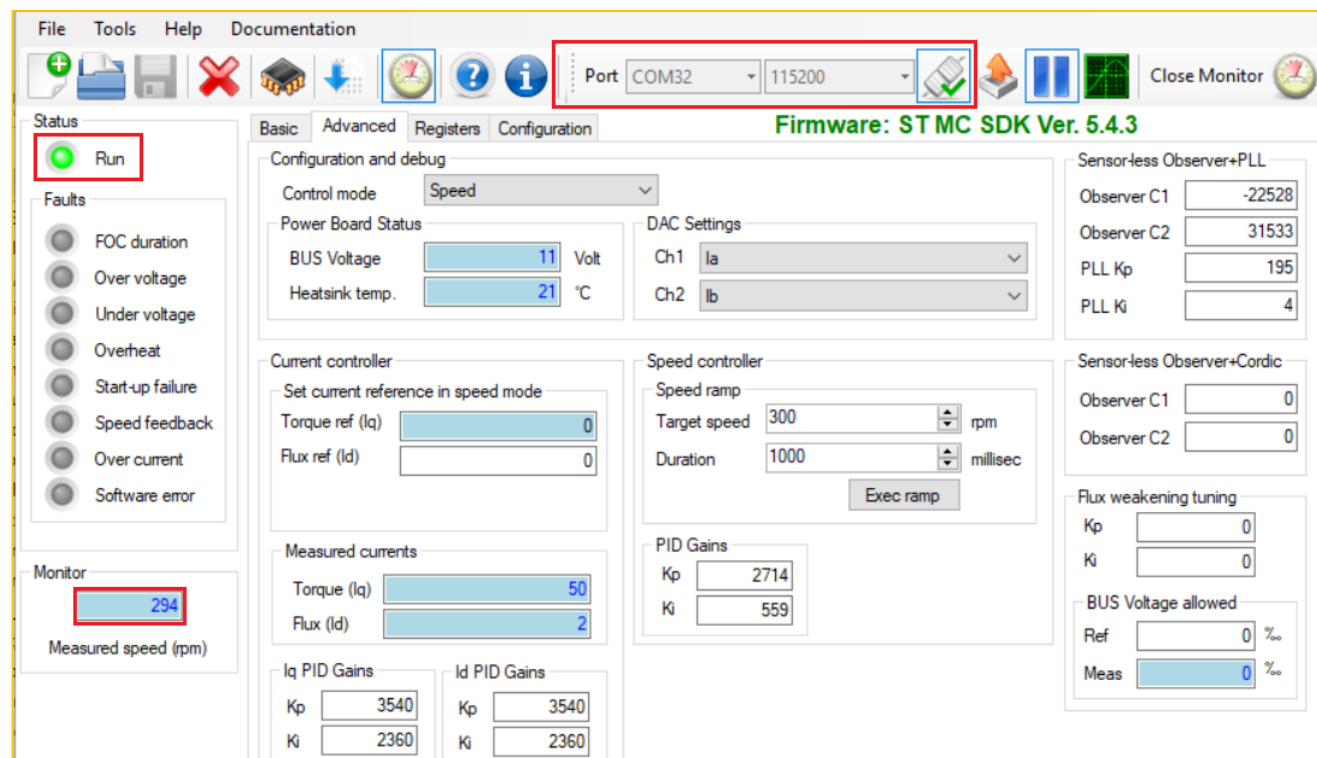


```
103
104  /* USER CODE END SysInit */
105
106  /* Initialize all configured
107  MX_GPIO_Init();
108  MX_ADC1_Init();
109  MX_ADC2_Init();
110  MX_CORDIC_Init();
111  MX_DAC1_Init();
112  MX_TIM1_Init();
113  MX_USART2_UART_Init();
114
115  /* Initialize interrupts */
116  MX_NVIC_Init();
117  /* USER CODE BEGIN 2 */
118  MX_MotorControl_Init();
119  /* USER CODE END 2 */
120
121  /* Infinite loop */
122  /* USER CODE BEGIN WHILE */
123  while (1)
124  {
```

验证结果

- 编译下载后验证电机是否正常工作

```
Build Output
Build target 'G474_MC'
compiling ntc_temperature_sensor.c...
linking...
Program Size: Code=32396 RO-data=784 RW-data=1904 ZI-data=1920
FromELF: creating hex file...
"G474_MC\G474_MC.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:14
Load "G474_MC\G474_MC.axf"
Erase Done.
Programming Done.
Verify OK.
Flash Load finished at 11:09:06
```



- 本篇章简单介绍了ST电机控制生态
- 基于STM32电机的FOC库应用操作
- 关于STM32G4上电机应用的外设ADC, TIMER, CORDIC具体应用
- 结合之前介绍的STM32G4技术篇章有助于电机控制具体应用

Thank you

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